



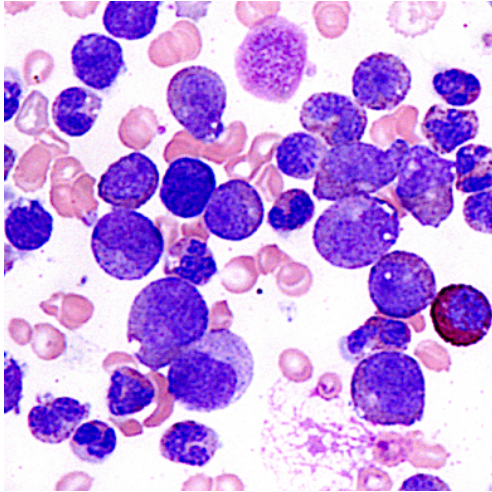
MEHLMANMEDICAL HY USMLE GENETICS

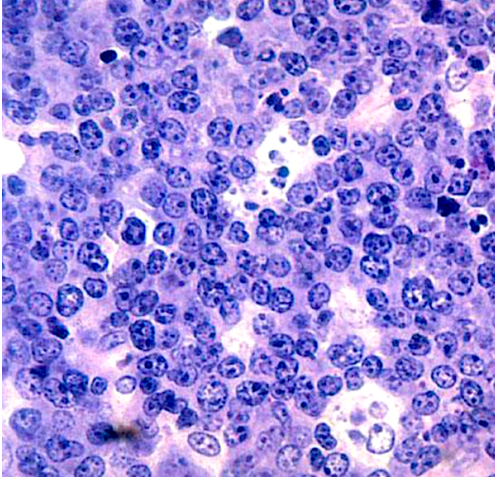


YouTube
@mehlmanmedical



Instagram
@mehlman_medical

HY Cancer Genes for USMLE	
<i>APC</i>	- Mutations cause Familial Adenomatous Polyposis (FAP); chromosome 5; AD. - Hundreds to thousands of polyps on colonoscopy; 100% cancer risk. - Answer on 2CK Peds form is total proctocolectomy in 18-year-old (presumably since 100% chance of cancer + too difficult to conservatively screen). - FAP + soft tissue (e.g., lipoma) or bone tumors (e.g., of the skull) = Gardner syndrome. - FAP + CNS tumors = Turcot syndrome.
<i>BCL-2</i>	- Overexpressed in follicular lymphoma (most common indolent non-Hodgkin lymphoma) as a t(14;18) translocation. - Codes for anti-apoptotic molecule. - Follicular lymphoma will present as waxing/waning painless lateral neck mass over 1-2 years in an adult. - USMLE might give research-type Q where <i>BCL-2</i> is overexpressed (unrelated to follicular lymphoma) and they ask what would be expected → answer = increased lifespan of population of cells (makes sense, since anti-apoptotic molecule).
<i>BCR-ABL</i>	- HY for chronic myelogenous leukemia (CML). - t(9;22) translocation of these two genes (aka Philadelphia chromosome) codes for a "fusion protein." - This fusion protein = an oncogenic tyrosine kinase. - CML is answer for leukemia when the Q gives you lots of myelo-sounding cells (i.e., myelocytes, promyelocytes, metamyelocytes). 4/5 Qs on USMLE that mention these cells are CML. Leukocyte ALP will be low.  - The blood smear for CML I refer to as a "motley mix," or a "soup." This is a buzzy image on NBME for CML. - Imatinib is Tx for CML. It can cause fluid retention / peripheral edema. - The MOA of imatinib = targets <i>BCR-ABL</i> tyrosine kinase.
<i>BRAF</i>	- Proto-oncogene that can be seen in some melanomas. - Codes for serine-threonine kinase.
<i>BRCA1/2</i>	- Breast cancer tumor suppressor genes. - NBME exam asks for what the gene does → answer = "recombinational ds-DNA repair." - Can also lead to gynecologic cancers in females or breast/testicular cancers in males. - Answer on an NBME exam for appropriate prophylaxis in patient who has confirmed mutation is "bilateral oophorectomy and mastectomy."
<i>c-KIT</i>	- Proto-oncogene; can be mutated in some gastrointestinal stromal tumors (GIST).
<i>c-MYC</i>	- Overexpressed in Burkitt lymphoma (a type of NHL) as a t(8;14) translocation. - Codes for a transcription factor. - Burkitt lymphoma will be mass of the jaw or abdomen. - Histo is buzzy for "starry sky" appearance, which is a basophilic (purple) background of B cells with scattered translucent macrophages.

	 - The macrophages are referred to as “tingible body” macrophages (correct, not tangible), where apoptosis occurs. This is asked on an NBME exam, where they have an arrow that points to one of the macrophages, and the answer is apoptosis.
<i>EGFR</i>	- Non-small cell lung cancer (adenocarcinoma) and glioblastoma multiforme. - Codes for EGFR tyrosine kinase. - Erlotinib asked on USMLE → targets EGFR tyrosine kinase.
<i>HER2/neu (ERBB2)</i>	- Tyrosine kinase expressed in some breast cancers. If (+) → poor prognostic indicator. - Trastuzumab (Herceptin) is a drug that targets <i>HER2/neu</i>. It can cause cardiotoxicity.
<i>JAK2</i>	- Can be activated in polycythemia vera, essential thrombocythosis, and myelofibrosis. - USMLE wants increased proliferation of hematopoietic stem cells, not pluripotent stem cells, for PV. Patient will have high RBCs, PLUS either high WBCs and/or platelets (i.e., 2-3 of the cell lines will be high in Qs, but RBCs are always high). Generalized pruritis after showers can be seen from basophilia. Hyperviscosity syndrome (i.e., blurry vision, headache, Raynaud) from high RBCs is treated with phlebotomy. Hydroxyurea can decrease recurrence of episodes. - Essential thrombocythosis will present in Qs as platelets over a million (NR 150-450,000). They will say there’s pain or discoloration in tips of fingers or give hyperviscosity-type findings. Bone marrow will show increased megakaryocytic proliferation (this latter finding will not be seen in reactive thrombocythosis, which is high platelets from infection). - Myelofibrosis will present with teardrop-shaped RBCs (dacryocytes) and/or “dry tap” on bone marrow aspiration. Massive splenomegaly is seen in basically all questions.
<i>KRAS</i>	- Proto-oncogene; codes for a GTPase. - The answer on USMLE for the first gene mutated in colonic polyps, prior to progression to overt colon cancer. - Colon cancer often develops as a result of <i>progressive</i> mutations, rather than one mutation straight-up. In other words, first <i>KRAS</i>, then <i>PTEN</i>, then <i>DCC</i>, then <i>TP53</i>. - If they tell you a colon cancer has metastasized and force you to choose a gene that’s mutated, go with <i>TP53</i> (codes for p53 protein). - If they tell you a polyp is seen and there is no evidence of invasion of the stalk, choose <i>KRAS</i>. This is on NBME exam.
<i>MEN1</i>	- Mutations cause Multiple Endocrine Neoplasia type I. - 3Ps → Pituitary tumor (e.g., prolactinoma), Parathyroid adenoma (or diffuse 4-gland hyperplasia), and Pancreatic tumor (e.g., gastrinoma; Zollinger-Ellison syndrome). - HY point is that NBME Qs need not give all findings within a MEN syndrome. For example, the Q can give gastrinoma alone and ask for the gene → <i>MEN1</i>.
<i>MSH2/6</i> <i>MLH1</i> <i>PMS2</i>	- Hereditary non-polyposis colorectal cancer (HNPCC). - Mismatch repair genes; mutations cause “microsatellite instability.” - Colonic polyps/cancer; also associated with gynecologic cancer.
<i>NF1</i>	- Neurofibromatosis type I; chromosome 17; AD.

	- Café au lait spots (hyperpigmented macules), axillary/groin freckling, neurofibromas (nerve sheath tumors presenting as nodules under the skin), weird CNS tumors (i.e., oligodendroglioma, ependymoma, meningioma), optic glioma (CN II tumor). - Classic disease that demonstrates variable expressivity, which means varying disease severity (I talk more about this stuff in my HY Genetics PDF).
<i>NF2</i>	- Neurofibromatosis type II; chromosome 22; AD. - Presents with acoustic schwannoma. Can be bilateral. - Meningioma.
<i>RB</i>	- Congenital retinoblastoma (i.e., leukocoria in 1-year-old) and osteosarcoma. - Tumor-suppressor gene; autosomal dominant (two hits required, but chance of second mutation is 100%, so AD not AR). - RB protein is normally in a complex with E2F, a transcription factor, repressing it and preventing cell cycle progression. CDK/Cyclin complexes then phosphorylate RB, releasing it from E2F. E2F then goes to the nucleus and transcribes genes → cell cycle progression. - NBME simply wants you to know that decreased RB phosphorylation is a wrong answer for what we'd expect in cancer cells.
<i>RET</i>	- Proto-oncogene → MEN 2A/2B. - Both MEN 2 syndromes have pheochromocytoma and medullary thyroid carcinoma. - MEN 2A has parathyroid adenoma or hyperplasia (same as MEN 1). - MEN 2B has Marfanoid body habitus and/or mucosal neuromas. - Same as with MEN 1, a HY point is that NBME Qs need not give all findings within a MEN syndrome. For instance, one NBME Q just gives medullary thyroid carcinoma alone, and the answer is <i>RET</i>. Another gives groin pain (urolithiasis due to hypercalcemia) + a neck tumor, and the answer is <i>RET</i>.
<i>TP53</i>	- Codes for p53 tumor-suppressor protein; halts the cell cycle in the setting of cellular damage so that DNA repair can occur. - Li-Fraumeni syndrome = congenital mutation in <i>TP53</i> leading to cancers of various organ systems. - HY answer on NBME for the gene mutated in cancer that has metastasized. - Pleiotropy = one gene has multiple effects; <i>TP53</i> is textbook example, since mutation can cause many unrelated cancers, such as pancreatic, ovarian, colon, etc.
<i>TSC1/2</i>	- Autosomal dominant; hamartin and tuberlin proteins. - Intellectual disability; periventricular nodules (tubers); adenoma sebaceum (aka angiofibromas, which are skin-colored/reddish papules on cheeks, nose, and in nasolabial folds); subungual fibromas (nailbed tumors); cardiac rhabdomyoma (ball-in-valve murmur that presents as diastolic rumble that attenuates with change of positioning); renal angiomyolipoma.
<i>VHL</i>	- Autosomal dominant; chromosome 3. - Renal cell carcinoma, PLUS retinal and/or cerebellar hemangioblastomas. - Pancreatic cysts can also be seen (on NBME). - The RCC need not be bilateral; don't confuse with angiomyolipoma of TSC. - New NBME Q shows picture of gross kidney lesion + tells you patient has cerebellar hemangioblastoma + retinal angioma; they ask what kind of kidney lesion they're showing → answer = "renal adenocarcinoma"; angioma is wrong answer.
<i>WT1</i>	- Can be isolated Wilms tumor; the answer on USMLE for kidney tumor in a kid almost always; presents as painless flank mass in 2-4-yr old. However Wilms tumor can also present as part of constellation syndromes: - Denys-Drash syndrome = gonadal agenesis + Wilms tumor. - WAGR syndrome = Wilms tumor, Aniridia, Genitourinary abnormalities, Retardation (can be associated with other genes as well, but just know that WT1 is associated). - Beckwith-Wiedemann syndrome = fetal macrosomia, macroglossia, hemihypertrophy, hypoglycemia, and Wilms tumor. Harder Q on 2CK Peds form gives big baby with hemihypertrophy + doesn't mention Wilms tumor; they ask what else could be seen → answer = hypoglycemia.

Other HY Genes for USMLE	
<i>ATP7B</i>	- Wilson disease; autosomal recessive; chromosome 13. - Copper overload. - Inability to secrete copper into bile from the liver. Copper is normally excreted by the body via secretion into bile. - ↑ urinary copper + ↓ serum ceruloplasmin (copper-binding protein in the blood; in the case of copper overload, body tries to minimize amount carried in blood). - Buzzy / pass-level detail is Keiser-Fleischer rings, which is copper deposited in the cornea of the eye. Vignette can give you what sounds like Wilson disease, and then the answer is “slit-lamp exam.” - Can cause ↑ LFTs with cirrhosis, hemolytic anemia, and Parkinsonism. - Copper deposits in basal ganglia, especially the putamen. - Parkinsonism in a young patient = Wilson until proven otherwise. - In old patient, Parkinsonism = Parkinson disease, normal pressure hydrocephalus, Lewy-body dementia, or progressive supranuclear palsy. - Treat with the copper chelator penicillamine.
<i>CFTR</i>	- Cystic fibrosis; chromosome 7; AR. - Codes for chloride channel. - Channel is normally located at cell surface; if mutated, it instead remains sequestered in the rough endoplasmic reticulum. - “Abnormal protein folding” is answer on NBME for result of <i>CFTR</i> mutations. - Delta F508 ($\Delta F508$) is most common mutation, which is deletion of phenylalanine 508. - CF is textbook example of allelic heterogeneity, which means many different mutations can cause the same disease. For this reason, most genotyping panels lack sensitivity, and sweat-chloride test showing >60 mEq/L is most diagnostic/accurate. A nasal test showing increased potential difference can also be performed. - Presents as child with chronic history of lung infections (<i>Pseudomonas</i> exceeds <i>S. aureus</i> after age 10). - Secretions in the alveoli and pancreatic ducts are inspissated (meaning desiccated / dried up within a lumen), making them sticky. This leads to exocrine pancreatic insufficiency, where enzymes can’t make it to the duodenum → fat-soluble vitamin malabsorption → NBME exams love vitamin E deficiency in CF in particular (presents as neuropathy). - Meconium ileus at birth; buzzy, but often not mentioned in questions. - Phenotypically normal siblings of affected children have 2/3 chance of being carrier (holds true for any AR disorder). - Ivacaftor/lumacaftor are newer treatments. They are known as potentiators/correctors, where the Cl^- channel structure, location, and function are improved.
<i>DMD</i>	- XR disorder caused by mutation in dystrophin. - Mutation results in disruption of α-/β-dystroglycan, which is required for proper internal cytoskeletal anchoring of the muscle cell to the extracellular matrix. - Presents with pseudohypertrophy, where muscles appear large but are replaced with fibroadipose tissue (connective tissue stromal cells). - Duchenne presents in a young boy who implements Gower maneuver to stand up (uses arms to walk up off the floor because leg muscles are weak). - Becker presents in adolescence or young adulthood (less severe form of Duchenne). - Duchenne is classically frameshift mutation; Becker is classically not frameshift.
<i>FBN1/2</i>	- Marfan syndrome; autosomal dominant; chromosome 15. - Codes for fibrillin, which is a glycoprotein that forms a sheath around elastin. - Tall, lanky body habitus with flat feet, chest wall abnormalities (i.e., pectus excavatum or carinatum), flat feet (pes planus), scoliosis, mitral valve prolapse (mid-systolic click), increased risk for aortic dissection (can retrograde propagate toward aortic root, causing root dilatation and aortic regurgitation [decrecendo diastolic murmur]). - Is not associated with berry/saccular aneurysms (unlike Ehlers-Danlos and ADPKD).
<i>FMR1</i>	- Fragile X; X-linked; caused by CGG trinucleotide repeat (TNR) expansion. - Results in hypermethylation of the gene and transcriptional silencing.

	- Presents as boy with large, everted ears; long, narrow jaw; macroorchidism; and intellectual disability. - Can sometimes be symptomatic in females if skewed X-inactivation (lyonization).
<i>FXN</i>	- Friedreich ataxia; Frataxin gene; GAA TNR expansion. - Presents with ataxia (you guessed it), scoliosis, cardiomyopathy, and decreased reflexes.
<i>G6PD</i>	- Glucose-6-phosphate dehydrogenase deficiency; X-linked recessive. - Presents as boy with hemolysis leading to jaundice (unconjugated hyperbilirubinemia) following oxidizing drug (e.g., sulfa, dapsone, primaquine). - Heinz bodies (denatured hemoglobin) and bite cells (degmacytes; partially phagocytosed RBCs due to removal of Heinz bodies) are seen.
<i>HFE</i>	- Hereditary hemochromatosis; autosomal recessive; chromosome 6. - Iron overload. - Mechanism USMLE wants is “increased intestinal iron absorption.” - Body has poor ability to excrete iron; occurs naturally via menses in women; otherwise there are minor losses via skin shedding. - Main iron regulation is via shutting off intestinal absorption; this is impaired in hemochromatosis. - Usually presents in adulthood in males first (because of menses in women). - Can present as “bronze diabetes” → hyperpigmentation due to hemosiderin deposition in skin + ↑ fasting sugars (iron deposition in tail of pancreas). - Miscellaneous other findings can be seen like infertility (iron deposition in hypothalamus, anterior pituitary, or gonads), cardiomyopathy, or arthritis (pseudogout). - Hereditary hemochromatosis, primary hyperparathyroidism, and hypothyroidism are 3 most important causes of pseudogout. I used to only discuss the former two with students over the years, but the latter shows up on a new 2CK NBME exam where they mention chondrocalcinosis (calcium deposition in cartilage). - USMLE wants you to know there is ↑ risk of hepatocellular carcinoma. There is easy NBME Q of hemochromatosis where they ask what patient is at increased risk for, and the answer is simply “hepatocellular carcinoma.” - Diagnose with ferritin >300 mg/dL. Transferrin saturation will also clearly be ↑. - It is exceedingly rare that ferritin is >300 in other conditions, however this can occur in lymphoma and leukemia, where ↑ ferritin is a poor prognostic marker in non-Hodgkin lymphoma. There is one NBME Q on 2CK where ferritin is 300 where it’s not hemochromatosis, but USMLE won’t play gotcha. - Treat with serial phlebotomy, not chelators. - Chelators such as deferoxamine or deferasirox are for secondary hemochromatosis due to transfusional siderosis (i.e., repeated blood transfusions that contain iron, for e.g., β-thalassemia major).
<i>HTT</i>	- Huntington disease; autosomal dominant; CAG TNR expansion on chromosome 4. - Presents as cognitive decline and choreoathetosis, usually in patient 30-40s. - Chorea = fast, purposeless, jerky movements. - Athetosis = slow, writing movements. - TNR disorders demonstrate anticipation, which means they become more severe and earlier-onset with each generation due to further expansion of the TNR, so the vignette can say the, e.g., 40-year-old patient had parent with similar symptoms appearing in his/her 50s. - If the USMLE asks you for which part of the brain is fucked up, choose caudate nucleus.
<i>NFKB1/2</i>	- Codes for NF-κB protein, which is involved in a myriad of cell-signaling processes. - For whatever magical reason, a repeated question across NBME exams wants you to know that “IκB releases NF-κB after undergoing phosphorylation.” - NF-κB is then free to go to the nucleus to upregulate transcription of 150+ genes. - Glucocorticoids (i.e., such as prednisone) partially exert their immunosuppressant effects by inhibiting NF-κB-mediated gene expression.
<i>PKD1/2</i>	- Autosomal dominant polycystic kidney disease. - The answer on USMLE if disease starts as an adult (i.e., 30s-40s). - Cysts are technically present early in life, but only become clinical as adult (i.e., ↑ BP and ↑ RFTs).

	- These patients have ↑ BP due to ↑ RAAS (compression of microvasculature of kidney due to enlarging cysts). - Can cause saccular (berry) aneurysms of the circle of Willis → risk for subarachnoid hemorrhage. - Highest yield point is that serial blood pressure checks are correct over circle of Willis MR angiogram screening. Latter is wrong answer on USMLE. MR angiogram screening of circle of Willis is only done when there is (+) family Hx of SAH or saccular aneurysms. - Step 1 NBME gives easy vignette of ADPKD, and then the answer is just “polycystin” as the protein that’s fucked up. Sounds obvious, but it’s asked so I’m mentioning it. - Cystic kidneys are part of “ciliopathies,” which is obscure term that refers to conditions where cilia are abnormal. Polycystin is a protein required for cilia function on renal epithelium. - Most common extra-renal location for cysts is the liver (85% by age 30).
<i>PKHD1</i>	- Autosomal recessive polycystic kidney disease. - The answer on USMLE for cystic kidneys in pediatrics. - Can be associated with hepatic fibrosis.

1. A 4-year-old boy is brought to the physician because of a chronic cough that is worse at night and productive of thick white sputum. He is at the 3rd percentile for weight and height. Physical examination shows clubbing of the fingers. Chest auscultation shows diffuse inspiratory crackles and hyperresonance. The patient has a healthy 6-year-old sister. Which of the following represents the likelihood of this patient's sister being a carrier of this condition?
- A) 1 of 1
 - B) 1 of 2
 - C) 1 of 3
 - D) 1 of 4
 - E) 2 of 3
 - F) There is no chance the patient's sister is a carrier

The correct answer is E.

Cystic fibrosis is an autosomal recessive condition on chromosome 7. There is a one-half chance each parent will pass his or her mutated allele to the offspring. In the following Punnett square, “A” refers to a non-diseased wild-type allele; “a” refers to a diseased allele.

2/3 chance of being carrier

AA	Aa
Aa	aa

2x Aa / 3 remaining possibilities

Since the sister is healthy (i.e., phenotypically normal), *we know she can't possibly be “aa,” so we can eliminate that.* This leaves us with only three possibilities: AA + 2(Aa).

$2(Aa) / 3 \text{ total possibilities} = 2/3 \text{ chance of being a carrier.}$

Bottom line: Phenotypically normal siblings of patients with AR disorders bear a 2/3 chance of being a carrier. This is an important genetics ratio you need to memorize for USMLE, as the exam will often ask harder questions, where knowing this ratio is a mere component of, rather than answer to, the question.

2. A 34-year-old woman presents to preconception counseling with her husband. Her family history is remarkable for a brother who died at the age of 3 from deficiency of α -L-iduronidase. Her husband's family history is unremarkable. If the frequency of disease in the general population is approximately 1 in 100,000 for this condition, what is the probability this couple will have a son with the disease?
- A) 1/300
 - B) 1/900
 - C) 1/1900
 - D) 1/4200
 - E) 1/7600

The correct answer is C.

First solve for carrier frequency in general population (husband's carrier risk):

Hardy-Weinberg equation: $1 = p^2 + 2pq + q^2$, where $p + q = 1$.

$$q^2 = 1/100,000$$

$$\text{So } q = \sqrt{1/100,000} = 1/316.$$

$$\text{So } p = 1 - (1/316) \approx 1.$$

$$2pq = 2 \times 1 \times 1/316 = 1/158 = \text{carrier frequency in general population.}$$

Next solve for chance the patient (mom) is a carrier:

Hurler syndrome is an AR disorder. As per question #1, there is a 2/3 a phenotypically normal patient is a carrier of an AR disorder if he or she has a sibling with the disorder.

2/3 chance of being carrier

AA	Aa
Aa	aa

2x Aa / 3 remaining possibilities

Now solve for the risk this couple would have a child who has the AR disease:

$$2/3 \text{ (chance mom is carrier)} \times 1/158 \text{ (chance dad is carrier)} \times 1/4 \text{ (chance of aa in Punnett square)} = 2/1896.$$

However the question asked what is the chance the couple will have a *son* with the disease, so now we multiply by 1/2 (chance of having a son). This would be the same for daughters because this is an AR disorder.

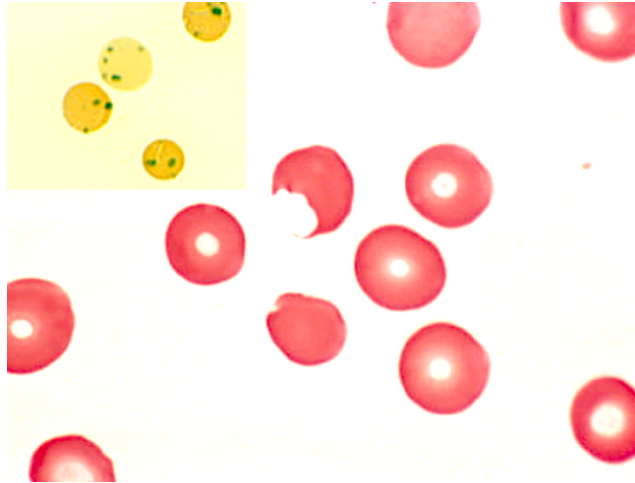
$$\text{So } 2/1896 \times 1/2 = \mathbf{1/1896} \approx 1/1900.$$

The USMLE will often give you conditions where q^2 is, e.g., 1/10,000, so q is easier to solve for (1/100), without the need for decimals.

3. An 8-year-old boy is brought to the physician by his mom for yellow discoloration of his eyes she first noticed this morning. He recently recovered from an upper respiratory tract infection. Serum studies show:

Hemoglobin	10.3 g/dL
Hematocrit	30.9%
Total bilirubin	4.0 mg/dL
Direct bilirubin	0.3 mg/dL

A peripheral blood smear is shown:



The boy's parents would like to have two more children. What is the probability that both children will carry the same diagnosis?

- A) 1 of 1
- B) 1 of 2
- C) 1 of 4
- D) 1 of 8
- E) 1 of 16
- F) 1 of 32
- G) 1 of 64
- H) Cannot be determined as this condition is usually sporadic *de novo*

The correct answer is E.

The diagnosis is glucose-6-phosphate dehydrogenase (G6PD) deficiency, an X-linked recessive disorder.

Unless in the rare event of heavily skewed lyonization (X-inactivation), only boys will have the disorder.

The patient has hemolysis precipitated by viral infection. The smear shows Heinz bodies (oxidized, precipitated hemoglobin) and bite cells (degmacocytes; partially phagocytosed RBCs).

The mother must be a carrier for the disorder (i.e., she has one X chromosome with the mutation).

The chance of her having another son is $1/2$. The chance of her passing the mutation-laden X-chromosome to this son is $1/2$. So the chance of having another son with the disease is $1/4$.

The chance therefore of two sons in a row with the disease would be $1/4 \times 1/4 = 1/16$ = the chance this couple's next two children will have G6PD deficiency.

4. A newborn male dies shortly after birth due to multiple organ system abnormalities. Autopsy shows an apparent transformation of lumbar vertebrae to a thoracic identity. Which of the following mechanisms relating to *HOX* gene may best explain this observation at the thoracic level?
- A) Alteration in transcription factor binding sites at the thoracic-lumbar vertebral junction
 - B) Expression of a *HOX* gene normally transcribed only caudally
 - C) Expression of a *HOX* gene normally transcribed only cranially
 - D) Insufficient expression of a *HOX* gene at the lumbar level
 - E) Insufficient expression of a *HOX* enhancer sequence at the thoracic level

The correct answer is C.

HOX genes are involved in the patterning of an embryo along the head-tail axis – i.e., they encode the characteristic positions of body parts, ensuring development at correct locations.

For example, lumbar vertebrae develop specifically at the lumbar level, and not at the sacral, thoracic, or cervical levels, because *HOX* genes ensure this specific type of developmental patterning.

If lumbar vertebrae are transformed into a thoracic identity – i.e., if vertebrae at the lumbar location have developed to appear like thoracic ones – then one possible mechanism is that thoracic *HOX* genes were inadvertently expressed at the lumbar location.

Since thoracic *HOX* genes are cranial to the lumbar ones, the transformation of lumbar vertebrae into thoracic ones means that genes normally expressed only superior (cranial) got expressed at an inferior (caudal) location.

Choice B would refer to the transformation of thoracic vertebrae into lumbar ones.

5. A 42-year-old woman presents to the physician with a 3-month history of progressive shortness of breath on exertion. She has a history of recurrent epistaxis since childhood. Echocardiogram shows an ejection fraction of 75%. A photograph of the patient's tongue is shown.



The patient's husband does not have a family history of the condition. What is the probability the patient and her husband will have a daughter who is a carrier?

- A) 0
- B) 1 of 1
- C) 1 of 2
- D) 1 of 3
- E) 1 of 4
- F) 1 of 6
- G) 1 of 8
- H) Cannot be determined without knowing the husband's carrier status

The correct answer is A.

The diagnosis is hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu), an autosomal dominant disorder.

The image shown is a telangiectasia of the tongue. The USMLE is also known to show an image of nailbed telangiectasias.

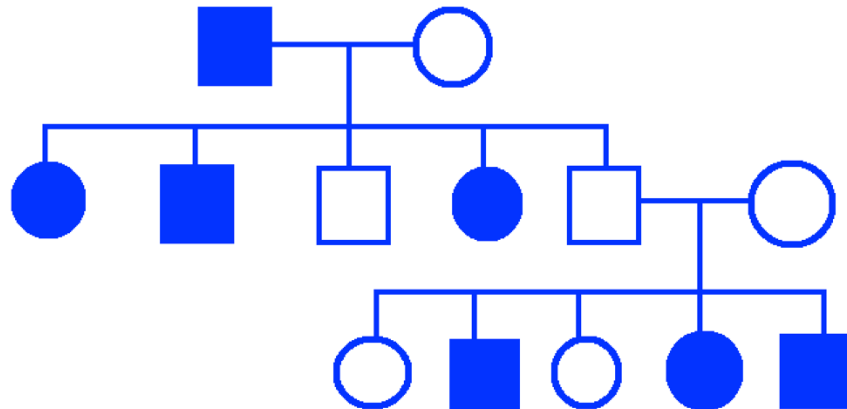
This patient has high-output cardiac failure secondary to a pulmonary arteriovenous fistula, which can present in adulthood. Telangiectasias of the nasal mucosa resulting in recurrent epistaxis since childhood are characteristic. Telangiectasias within the GI tract can cause GI bleeding and anemia.

Autosomal dominant disorders do not have a carrier state, since only one mutation results in disease phenotype. Carrier states are only possible in AR and XR disorders.

Even if a disease has incomplete penetrance (i.e., not all patients with diseased genotype bear the diseased phenotype), a patient with a dominant mutation would still not be considered a carrier. He or she would be said to have an AD disorder, albeit with reduced penetrance.

The husband cannot be a carrier, since HHT is an AD disorder and he does not have family history.

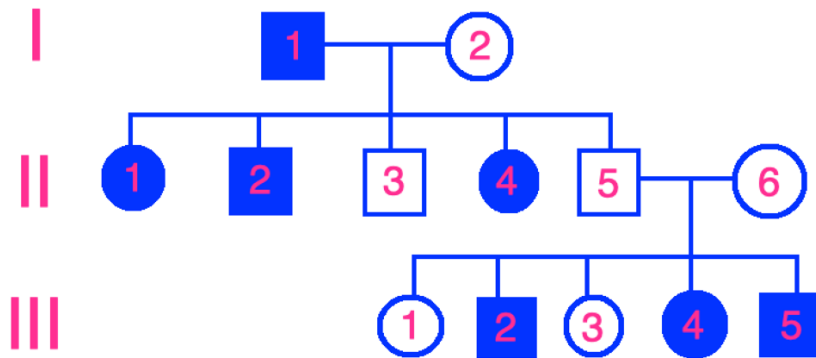
6. A 39-year-old man and his wife have two sons with history of retinoblastoma and a daughter with osteosarcoma. Neither the patient nor his wife have history of health problems. The patient is identified as II-5 on the pedigree.



Which of the following best explains the reason for the patient's health?

- A) Germline mosaicism
- B) Heteroplasmy
- C) Incomplete penetrance
- D) Somatic mosaicism
- E) Variable expressivity

The correct answer is C.



This patient's (II-5) family has congenital retinoblastoma. Retinoblastoma in infancy and/or osteosarcoma later in life are most characteristic of the condition.

Congenital retinoblastoma is an autosomal dominant disorder with incomplete penetrance.

Incomplete penetrance is when not everyone with the diseased genotype bears the diseased phenotype (i.e., not everyone with the disease expresses it); this is sometimes referred to as “skipping a generation,” and also applies to the *BRCA* genes.

Variable expressivity, in contrast, implies 100% penetrance – i.e., everyone with the disease expresses it in some form. The difference is the mere severity in which family members manifest the condition. *NF1* is a classic example, where everyone with the disease has it, but whilst one family member might only present with axillary and groin freckling, another might develop pheochromocytoma and optic glioma. **If the NBME/USMLE gives you a pedigree that looks like incomplete penetrance (as with this question), but the diagnosis in the family is *NF1*, choose variable expressivity, not incomplete penetrance.** A patient with minimal disease severity in *NF1* might not report having it.

Heteroplasmy is a phenomenon seen in mitochondrial disorders, an inheritance pattern characterized by strict maternal transmission (i.e., mothers transmit disease to sons and daughters; fathers never transmit disease), where varying amounts of diseased mitochondria are passed to the next generation, resulting in offspring with disease of varying severity. This is due to the mother's oocytes carrying thousands of copies of mitochondrial DNA, where the oocytes differ in the ratios of diseased mitochondria they contain. Do not confuse this with variable expressivity, which instead refers to non-mitochondrial disorders, where a chromosomal mutation can result in varying severity of disease.

Somatic mosaicism is when only a percentage of an individual's “body cells” (somatic = body) contain a diseased genotype. This results from a post-fertilization mitotic error, where the mutation results in only some of the organism's cells being affected by disease. Classic examples for USMLE would be mosaic Turner or Down, where the answer is “post-fertilization mitotic error.” Sturge-Weber is also an example of somatic mosaicism.

Germline (gonadal) mosaicism is when the spermatogonia or oogonia of one of the parents carry a mutation, but the somatic (body) cells do not. The result is that offspring can have an AD disorder despite the parents not being affected. Germline mosaicism is the answer classically for achondroplasia – i.e., the parents are normal but the child has the disease. The child will have 100% of his or her body cells affected because the original mutation was in the spermatogonia or oogonia of the parent, rather than occurring post-mitotically, as with somatic mosaicism. If the USMLE does not give achondroplasia, they will say two phenotypically normal parents have two children with an AD disorder, implying there is a germline mutation in one of the parents (usually the father).

7. A 2-year-old boy is brought to the pediatrician by his parents for a follow-up appointment. He has a history since birth of recurrent *Staph*, *Candida*, and viral infections. Examination shows scanty lymph nodes and tonsils and absent thymic shadow. Molecular analysis shows deficiency of adenosine deaminase. The boy's parents are planning to have more children. What is the probability that they will have a daughter with this molecular defect?
- A) Zero percent, since this condition is X-linked recessive
 - B) 1 in 2, since this condition is autosomal dominant
 - C) 1 in 2, since this condition is autosomal recessive
 - D) 1 in 4, since this condition is autosomal dominant
 - E) 1 in 4, since this condition is autosomal recessive
 - F) 1 in 8, since this condition is autosomal dominant
 - G) 1 in 8, since this condition is autosomal recessive
 - H) 2 in 3, since this condition is autosomal dominant
 - I) 2 in 3, since this condition is autosomal recessive

The correct answer is G.

The diagnosis is severe combined immunodeficiency (SCID). There are two main types of SCID:

- XR: common gamma chain mutation, or IL-2 receptor deficiency
- AR: adenosine deaminase (ADA) deficiency

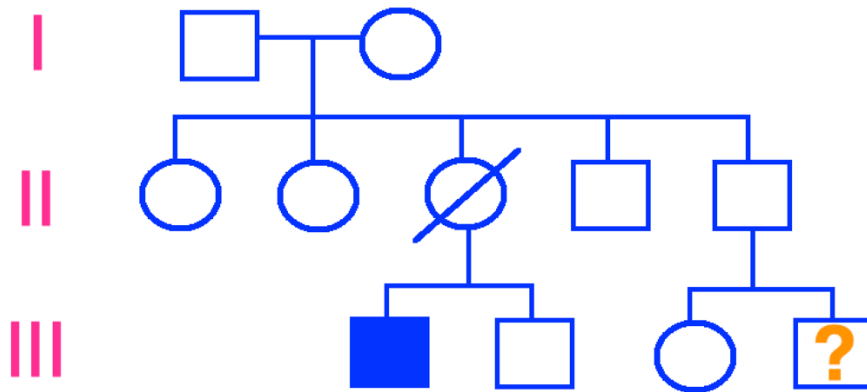
Since the patient has ADA deficiency, we know he has the AR, not XR, type of SCID.

This means there is a 1/4 chance the couple will have another child with AR SCID, and a 1/8 chance they will have a daughter with AR SCID (i.e., 1/4 chance of next child with SCID x 1/2 chance of having a daughter).

The ratio 2/3 refers to the probability that a phenotypically normal sibling of a patient with an AR disorder is a carrier for the disease.

If the patient in the vignette had common gamma chain mutation or IL-2 receptor deficiency (both XR SCID), then the answer would be zero percent.

8. A child (III-1 in the pedigree) has a history of a condition characterized by self-mutilation, aggressive behavior, and orange-red, sandy crystals in the diaper.



Assuming the mating partner of II-5 is genotypically normal, what is the probability that individual III-4 develops a similar condition?

- A) Zero percent
- B) 1 in 2
- C) 1 in 4
- D) 1 in 8
- E) 2 in 3
- F) Cannot be determined without carrier status of mating partner of II-5

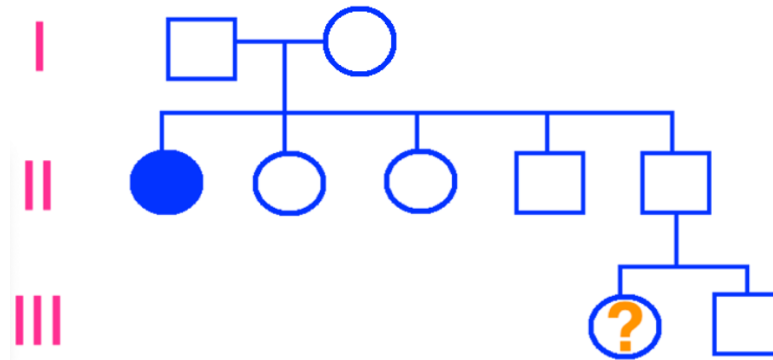
The correct answer is A.

The diagnosis is Lesch-Nyhan syndrome, an X-linked recessive disorder, caused by deficiency of the enzyme hypoxanthine-guanine phosphoribosyltransferase (HGPRT). This enzyme is required for the recycling of purines. Deficiency leads to gout due to overproduction of uric acid. Vignettes will give young boys who exhibit self-mutilation and red-orange sand-like crystals in their diapers.

XR diseases cannot be transmitted from father to son, so even if II-5 in the pedigree had the condition, he wouldn't be able to transfer it anyway.

If we consider daughter III-3, there is also a zero percent chance she would be a carrier for the condition as well, since II-5 does not have the disease.

9. A 54-year-old woman (II-1) presents to the cardiologist for a follow-up examination. She has an ejection fraction of 35% and a dilated cardiac silhouette. She has chronic joint pain of her hands and an occasional sore knee managed with indomethacin. Fasting blood glucose is 130 mg/dL.



If the disease frequency of this condition is $1/400$ in the general population, what is the probability that III-1 has the disease?

- A) $3/200$
- B) $3/400$
- C) $8/305$
- D) $15/304$
- E) $19/1200$
- F) $21/1559$

The correct answer is E.

The diagnosis is hereditary hemochromatosis, an autosomal recessive condition, caused by mutations in the *HFE* gene on chromosome 6. This results in iron overload caused by an inability to shut off iron absorption in the duodenum.

Since individual II-1 has the disease, we know that I-1 is a carrier. In addition, we also know that phenotypically normal siblings of individuals with AR disorders have a 2/3 chance of being a carrier, since “aa” is eliminated from the Punnett square (because they don’t have the disease). This means there is a 2/3 chance II-5 is a carrier.

Now we need to calculate the probability that the mating partner of II-5 is a carrier. Using our annoying as fuck Hardy-Weinberg equation:

$1 = p^2 + 2pq + q^2$, where $q^2 = 1/400$, and $q = 1/20$. Since $1 = p + q$, $p = 19/20$.

$2pq$ (carrier frequency) therefore $= 2(19/20)(1/20) = 38/400 = 19/200$ in general population.

Now we calculate the chance that III-1 has the disease:

$2/3$ (chance II-5 father is carrier) $\times$ $19/200$ (chance mother is carrier from general population) $\times$ $1/4$ chance of “aa” within the Punnett square $= 38/2400 = 19/1200$.

“Bronze” diabetes is a classic presentation for this condition, characterized by hyperpigmentation of the skin (due to hemosiderin deposition), diabetes mellitus (due to iron deposition in the tail of the pancreas), and various other findings, such as pseudogout (calcium pyrophosphate deposition disease), cardiomyopathy (dilated or restrictive), and infertility (due to hypothalamic or testicular iron deposition).

This presentation is slightly harder because hyperpigmentation is omitted from the stem, but the patient has dilated cardiomyopathy, pseudogout, and diabetes mellitus.

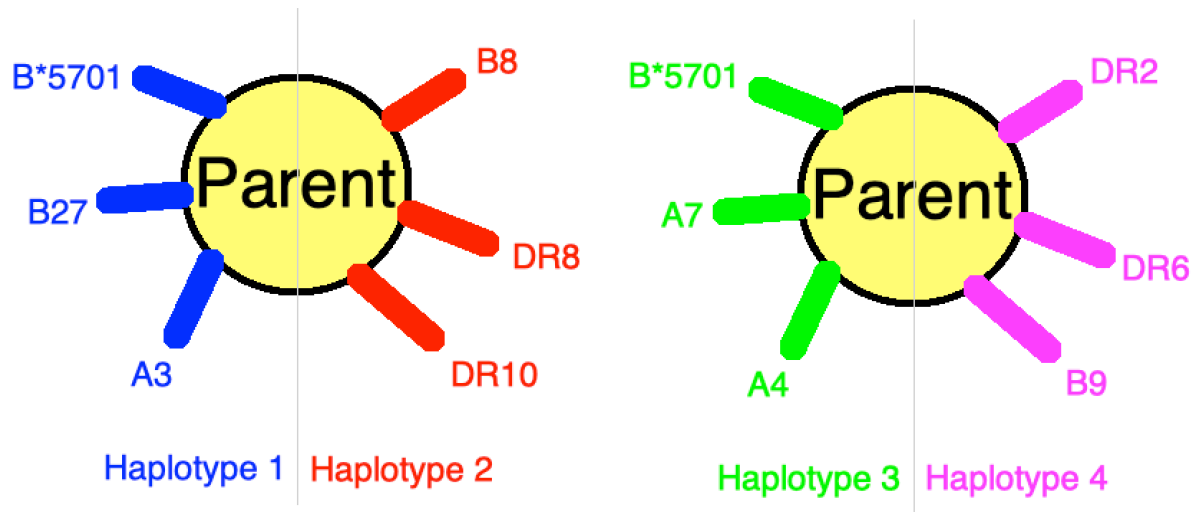
The two biggest risk factors for pseudogout are hereditary hemochromatosis and primary hyperparathyroidism. Pseudogout usually presents as either a monoarthritis of a large joint, such as the knee, or as a polyarthritis resembling osteoarthritis (OA). If the vignette gives you, e.g., Bouchard or Heberden nodes in a patient with hemochromatosis or primary hyperparathyroidism, the diagnosis is pseudogout, not OA.

Some students will ask, “But if we are forced to choose which cardiomyopathy it is, dilated or restrictive, which one is it?” The answer is: whichever one they give you, since the presentations are disparate. DCM presents with S3, enlarged silhouette, and decreased ejection fraction. RCM presents with S4, JVD, a normal silhouette, and normal ejection fraction.

10. A 36-year-old woman with HIV is undergoes HLA screening prior to commencement of abacavir as part of her anti-retroviral therapy. Genetic analysis confirms the homozygous presence of HLA B*5701 alleles. What is the likelihood this patient's brother is also homozygous for the same alleles?
- A) Impossible to determine since HLA inheritance is polygenic
 - B) 0%
 - C) 25%
 - D) 33%
 - E) 50%
 - F) 67%
 - G) 75%
 - H) 100%

The correct answer is C.

HLA genes are inherited as haplotypes, which are a set of alleles inherited from each parent. Each parent will have two haplotypes.



Possible offspring combinations

- Haplotype 1 Haplotype 3
- Haplotype 1 Haplotype 4
- Haplotype 2 Haplotype 3
- Haplotype 2 Haplotype 4

The HLA B*5701 allele would be carried within one of the haplotypes of each parent. The patient is homozygous for the HLA B*5701 allele. This means she has the **haplotype 1 / haplotype 3** combination. Since there are only four combinations, there is a 1/4 (25%) chance her brother would also have homozygous HLA B*5701 alleles.

Abacavir is known to cause a hypersensitivity-like reaction in patients with HLA B*5701 positivity. Screening for this HLA haplotype is required prior to commencement of abacavir.

11. A 21-year-old woman presents to the physician for a two-month history of weakness, headache, and blurry vision. Serum studies show a hemoglobin of 18.5 g/dL. Serum calcium is 10.6 mg/dL. Urinalysis shows 1+ RBCs. CT of the abdomen confirms the diagnosis. In addition, cysts are seen within the pancreas. Which of the following is the most likely mode of inheritance of this condition?
- A) Autosomal dominant
 - B) Autosomal recessive
 - C) Mitochondrial
 - D) X-linked dominant
 - E) X-linked recessive

The correct answer is A.

The diagnosis is von Hippel-Lindau (VHL), an autosomal dominant disorder on chromosome 3, which results in failure of degradation of hypoxia-inducible factor (HIF).

VHL is characterized by renal cell carcinoma (can become bilateral), pancreatic cysts, and cerebellar and retinal hemangioblastomas.

Renal cell carcinoma can present with polycythemia (due to EPO secretion) and hypercalcemia (paraneoplastic PTHrp secretion; similar to squamous cell carcinoma of the lung, yes).

Blurry vision and headache can be a result of hyperviscosity syndrome due to the high hemoglobin. The former could also be seen in retinal hemangioblastoma.

VHL is one of the phakomatoses (neurocutaneous disorders). The other high-yield ones for USMLE are neurofibromatosis 1 and 2 (NF1, NF2), tuberous sclerosis (TSC), and Sturge-Weber.

NF1, NF2, TSC, and VHL are all autosomal dominant. Sturge-Weber is not inherited and is a somatic mosaicism disorder (post-fertilization mitotic error).

NF1 (chromosome 17) is characterized by neurofibromas, axillary and groin freckling, café au lait spots (hyperpigmented macules), pheochromocytoma, optic glioma, and “weird” CNS tumors such as oligodendroglioma and ependymoma.

NF2 (chromosome 22) presents as acoustic schwannoma (can be bilateral) and meningioma.

TSC (chromosomes 9 and 16; coding for hamartin and tuberlin proteins, respectively) presents as periventricular nodules (tubers), adenoma sebaceum (angiofibromas; reddish-peach-colored papules in the nasolabial folds and on the bridge of the nose), cardiac rhabdomyoma, subungual fibromas, lymphangioleiomyomatosis, renal angiomyolipoma, ashleaf spots (hypopigmented macules), and shagreen patches (hyperpigmented velvety lesions). Vignettes will often be children presenting with seizures (i.e., writing movements during sleep) due to the tubers.

Sturge-Weber presents with Port wine stain birthmark (nevus flammeus; but can also present as violaceous papules in a temporal distribution), glaucoma, leptomeningeal angiomas, and seizures.

12. A 22-year-old woman presents to the physician for a two-month history of fatigue, mild cognitive degeneration, and clumsiness. Physical examination shows scleral icterus and a resting tremor of her left hand. Serum studies show elevated hepatic transaminases. Hemoglobin is 9.5 g/dL. Blood smear shows spherocytes. Which of the following is the most likely mode of inheritance of this condition?
- A) Autosomal dominant
 - B) Autosomal recessive
 - C) Mitochondrial
 - D) X-linked dominant
 - E) X-linked recessive

The correct answer is B.

The diagnosis is Wilson disease, an autosomal recessive condition on chromosome 13, characterized by an inability to excrete copper into bile and resultant copper overload.

Patients have increased urinary copper and decreased serum ceruloplasmin (copper-binding protein). The latter results in increased free serum copper that accumulates within RBCs, leading to hemolysis and spherocytosis.

Spherocytes are not limited to hereditary spherocytosis (which is autosomal dominant). The USMLE wants you to know they are frequently seen in the setting of drug- and infection-induced hemolysis. They are also seen in hemolytic anemia in Wilson disease. (Should be noted tangentially that they are *not* seen in G6PD deficiency; bite cells are seen instead)

Wilson disease can present as Parkinsonism in a young patient (i.e., unilateral resting tremor, bradykinesia, shuffling gait, etc.), and hence is known as a Parkinson-plus disorder (i.e., appears like Parkinson disease but is not). This is due to copper deposition within the basal ganglia. Other HY Parkinson-plus syndromes for USMLE are progressive supranuclear palsy, normal pressure hydrocephalus, Lewy body dementia, MPTP-induced Parkinsonism, and antipsychotic-induced Parkinsonism.

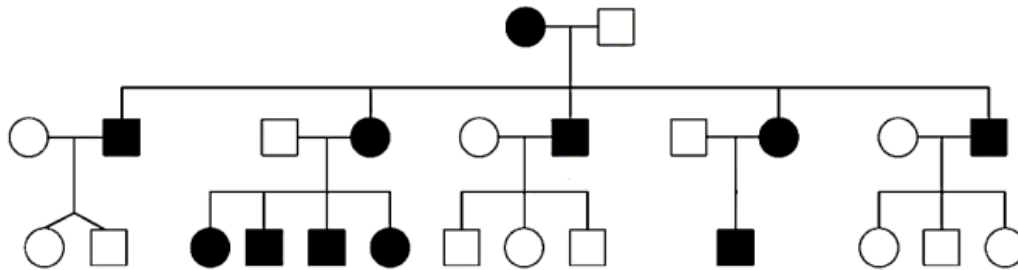
Liver disease and cirrhosis are also important findings in Wilson. LFTs may be variable depending on stage of liver failure. Cirrhotic patients will have elevated prothrombin time (PT).

Kayser-Fleischer rings are often too easy and omitted from USMLE vignettes. These are copper rings seen with Descemet membrane of the cornea. The USMLE might give you a vignette of Wilson disease, and then the answer is, "Perform a slit-lamp examination."

Treatment of Wilson disease is with penicillamine, a copper chelator. Trientine and zinc supplements may also be used.

Bottom line: The heavy metal disorders – Wilson and hemochromatosis – are autosomal recessive.

13. A 15-year-old boy has a 4-month history of progressive shortness of breath with exertion, muscle weakness, and worsening exercise tolerance. Vitals are within normal limits. Physical examination shows bilateral ptosis. Muscle strength is 3/5 in all extremities. A pedigree is shown. The patient is represented by III-10.



Which of the following best explains the combination of findings in this patient?

- A) Allelic heterogeneity
- B) Anticipation
- C) Chimerism
- D) Complete penetrance
- E) Heteroplasmy
- F) Pleiotropy

The correct answer is E.

This patient has a mitochondrial disorder. This is evidenced by the classic constellation of lactic acidosis, hypotonia, and ear and/or eye problem.

Mitochondrial inheritance is strictly maternal – i.e., only females can pass disease to offspring; males cannot transmit disease.

This is because the mitochondrial genes are separate from the chromosomal ones. The oocytes of the female contain thousands of copies mitochondrial genes, which are the source of mitochondrial DNA in the resultant zygote. The sperm does not contribute mitochondrial DNA to the zygote.

Each oocyte contains a different proportion/fraction of diseased mitochondria. Depending on the fraction of diseased versus normal mitochondrial DNA, the resultant offspring will bear varying disease severity. This is referred to as heteroplasmy.

Allelic heterogeneity refers to different alleles resulting in the same disease. Cystic fibrosis is a classic example, where a large number of possible mutations within the *CFTR* gene are known to cause disease, with $\Delta F508$ being most common. The Sweat chloride test is more accurate than genotyping for CF because common *CFTR* screening panels will only detect upwards of 96% of likely mutations. Sweat chloride test, however, will be positive in all patients with CF.

Anticipation refers to the propensity for trinucleotide repeat (TNR) disorders presenting earlier and more severe with each successive generation. This is due to an expansion of the number of repeats with each generation. For example, Huntington disease may present in a male at age 36 who has 55 repeats, whereas in his mother it presented at age 44, and she only had 48 repeats (40 or more CAG cause disease).

Chimerism is a phenomenon seen in some animal species, i.e., mice and cats, where more than two parents contribute DNA to the offspring. This is due to fusion of embryos. Chimerism is not seen in humans.

Complete penetrance is when 100% of those with a diseased genotype go on to develop the diseased phenotype. Incomplete penetrance is “skipping a generation,” where some individuals with a diseased genotype do not develop the diseased phenotype. Hereditary hemorrhagic telangiectasia, congenital retinoblastoma, and *BRCA* mutations are classic examples of incomplete penetrance.

Pleiotropy is when one gene results in many phenotypic effects – e.g., *TP53* mutations resulting in Li-Fraumeni syndrome can result in many different types of cancers (e.g., pancreatic, colon). Another example is phenylketonuria (PKU), where a mutation in phenylalanine hydroxylase can result in partial albinism, mental retardation, and musty body odor.

14. A 24-year-old woman presents to the physician with an ongoing history of heavy menses and epistaxis. Serum studies show:

Platelet count:	200,000/ μ L
Platelet aggregation studies:	Normal
Prothrombin time:	12 seconds
Partial thromboplastin time:	38 seconds
Bleeding time:	9 minutes
Ristocetin assay:	Suppressed

Which of the following is the most likely diagnosis?

- A) Glanzmann thrombasthenia
- B) Hemophilia A
- C) Hemophilia B
- D) Idiopathic thrombocytopenic purpura
- E) Vitamin K deficiency
- F) Von Willebrand disease

The correct answer is F.

Student says, “Wait, how is this a genetics question. I’m confused.” The USMLE will sometimes weave genetics knowledge into questions rather than asking you for inheritance patterns directly. This type of question is one of the highest yield applications of inheritance patterns.

Von Willebrand disease (vWD) is autosomal dominant. Hemophilia A and B are both X-linked recessive. Since this vignette is a female patient, even if you aren’t sure of the hematologic diagnosis, if you know the inheritance patterns, you can instantly eliminate the hemophilias because the patient is not male.

Von Willebrand disease will almost always present as a combination of one clotting factor problem (i.e., menorrhagia, excessive bleeding with tooth extraction) and one platelet problem (petechiae, purpura, ecchymoses, epistaxis). If the vignette unexpectedly gives you, e.g., only menorrhagia, you will be able to diagnose definitively based on the laboratory findings.

Von Willebrand factor (vWF) helps enable platelet *adhesion* by bridging glycoprotein Ib on platelets to vascular endothelium. The ristocetin assay will be abnormal in patients with impaired platelet *adhesion*. This means vWD and Bernard-Soulier disease (deficiency of GpIb) both have abnormal ristocetin assays.

Von Willebrand factor plays no role in platelet aggregation, where Gp IIb/IIIa on adjacent platelets are bridged together by fibrinogen. This means platelet aggregation studies are normal in vWD.

Bleeding time (normally 2-7 minutes) is increased if there is a platelet problem. Since platelet adhesion is disrupted in vWD, bleeding time is always increased.

Von Willebrand factor has an auxiliary role in helping to stabilize factor VIII in plasma. This means about half the time partial thromboplastin time (aPTT) will be increased. Normal range is 25-40 seconds. Since this stabilization of factor VIII is not the primary role of vWF, *aPTT is normal in about half of USMLE questions*. Students normally fuck this up, as they erroneously think aPTT is always increased in addition to BT. But aPTT is increased in only about half of questions.

Prothrombin time (PT) is normal in vWD because factor VIII is in the intrinsic, not the extrinsic, pathway.

Platelet count is normal in vWD because there is no mechanism for platelet destruction. There is merely impaired binding of Gp Ib on platelets to vascular endothelium.

Hemophilia A and B are deficiency of factors VIII and IX, respectively. The textbook vignette will be a school-age male with hemarthrosis. aPTT will be elevated. BT and PT are normal.

Glanzmann thrombasthenia is deficiency of glycoprotein IIb/IIIa on platelets, resulting in impaired aggregation. The ristocetin assay is normal in Glanzmann thrombasthenia.

Idiopathic thrombocytopenic purpura (ITP; aka immune thrombocytopenia purpura) is an isolated thrombocytopenia caused by antibodies against Gp IIb/IIIa on platelets. The classic presentation is a school-age kid or a woman 30s-40s with epistaxis or random bruising. Only BT is increased, and platelet count is always down.

Vitamin K deficiency presents with normal bleeding time, and increased PT and aPTT. Vitamin K does not affect platelets. It is a cofactor for the enzyme gamma-glutamyl carboxylase. Deficiency leads to impaired clotting due to inability to activate factors II, VII, IX, and X (as well as anti-clotting proteins C and S).

15. A 7-year-old girl with happy personality and mental retardation undergoes karyotypic analysis that does not reveal a deletion of the expected *UBE3A* gene on chromosome 15. Which of the following findings via fluorescent *in situ* hybridization is most likely to confirm the diagnosis in this child?
- A) Maternal imprinting
 - B) Maternal uniparental disomy
 - C) Paternal imprinting
 - D) Paternal uniparental disomy

The correct answer is D.

The diagnosis is Angelman syndrome (AS; aka “happy puppet” syndrome), caused usually by maternal mutation or deletion of the paternally imprinted *UBE3A* gene on chromosome 15. It can also be caused by paternal uniparental disomy.

Imprinting is a normal genomic-wide process via which certain genes demonstrate a silencing of one allele from one parent.

Maternal imprinting = allele coming from mom is silenced; therefore, only dad’s allele is expressed.

Paternal imprinting = allele coming from dad is silenced; therefore, only mom’s allele is expressed.

The *UBE3A* gene associated with Angelman syndrome is normally paternally imprinted, meaning even though two *UBE3A* alleles are inherited (one from each parent), only mom’s is normally expressed. If there is a maternal deletion or mutation, then even if the allele received from dad is normal, because dad’s allele is silenced, disease phenotype occurs because the child can only express the mutated/deleted maternal one.

“Mom is not an Angel” is the mnemonic for Angelman syndrome. Mom is not an angel because her gene is deleted (and the one received from dad is silenced).

This patient, however, was found to not have a deletion on karyotypic analysis. This means that maternal deletion/mutation was not the mechanism for disease here.

Uniparental disomy is when both alleles are erroneously inherited from one parent. Angelman syndrome can also be caused by paternal uniparental disomy, where both alleles are inherited from dad (but are silenced).

If both alleles are inherited from dad, even if they are both normal/not mutated, because both coming from dad would be silenced, this leaves no maternal allele that can be expressed, therefore resulting in disease.

AS = paternal imprinting + maternal gene deletion, OR paternal uniparental disomy.

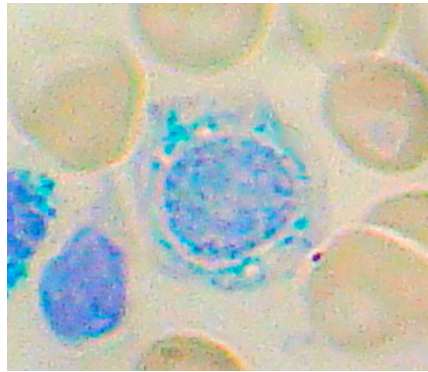
Prader-Willi syndrome (PWS) is another condition that demonstrates similar disease mechanisms as Angelman syndrome. PWS is characterized by hyperphagia (resulting in obesity) and mental retardation. PWS does not involve *UBE3A*, but can be caused by many genes on chromosome 15.

“Willi hates his dad” because dad’s gene is deleted/mutated (and mom’s is imprinted/silenced).

PWS can also be caused by maternal uniparental disomy, where both alleles are inherited from mom (but are silenced).

PWS = maternal imprinting + paternal gene deletion, OR maternal uniparental disomy.

16. A 4-year-old boy with a congenital enzyme deficiency is brought to the pediatrician for a follow-up examination. The disease frequency is approximately 1 in 200,000 in males. A bone marrow aspirate is shown.



Which of the following is the expected prevalence of heterozygous females for this condition?

- A) 1 in 20,000
- B) 1 in 40,000
- C) 1 in 50,000
- D) 1 in 100,000
- E) 1 in 200,000
- F) 1 in 400,000

The correct answer is D.

Congenital sideroblastic anemia is an X-linked recessive disorder. The image shows a ringed sideroblast on Prussian blue staining.

For XR disorders, disease frequency in males = q . In females, disease frequency = q^2 .

If the frequency in males is $1/200,000 = q$, then disease frequency in females = $(1/200,000)^2$.

This question is not asking for the disease frequency in females, however. It is asking for carrier (heterozygote) frequency.

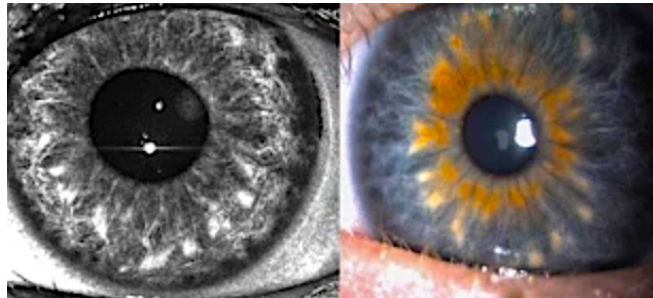
Using Hardy-Weinberg: $1 = p^2 + 2pq + q^2 = (p + q)$, since $q = 1/200,000$, $p = 199,999/200,000 \approx 1$.

$2pq = 2 \times 1 \times 1/200,000 = 1/100,000 =$ carrier frequency in females. This only applies to females because males cannot be carriers for XR disorders. So carrier frequency in females = carrier frequency in the general population for XR disorders.

In summary, for XR disorders:

- Disease frequency in males = q
- Disease frequency in females = q^2
- Carrier (heterozygote) frequency in females = $2pq$

17. A 3-year-old girl is evaluated for conductive hearing loss due to Eustachian tube dysfunction and middle ear fluid collections. Physical examination shows slanted palpebral fissures and single palmar creases. One of the patient's eyes is shown both under visible and near-infrared wavelengths. Fluorescent *in situ* hybridization shows cellular uniformity of karyotype.



The girl's mother is currently 17 years old. Which of the following is the most likely mechanism for the patient's condition?

- A) Deletion within *WT1* gene
- B) Duplication within *WT2* gene
- C) Meiotic nondisjunction
- D) Post-fertilization mitotic error
- E) Translocation

The correct answer is E.

The diagnosis is Down syndrome (trisomy 21). The images show Brushfield spots, which are connective tissue deposits seen within the irides of some patients with Down syndrome.

Eustachian tube dysfunction and middle ear fluid collection (“glue ear”) leading to conductive hearing loss is common in Down syndrome. It should be noted that single palmar crease, although frequently mentioned in Down syndrome questions, is non-specific. There is a 2CK psych form that has single palmar crease mentioned in a fetal alcohol syndrome question.

Approximately 95% of cases of Down syndrome are due to meiotic nondisjunction, most often secondary to advanced maternal age (i.e., 40 or older).

The patient’s mother was only 14 years old when she gave birth. The implication is that an alternative mechanism is most likely.

Approximately 4% of cases are due to Robertsonian translocation; 1% are due to mosaicism.

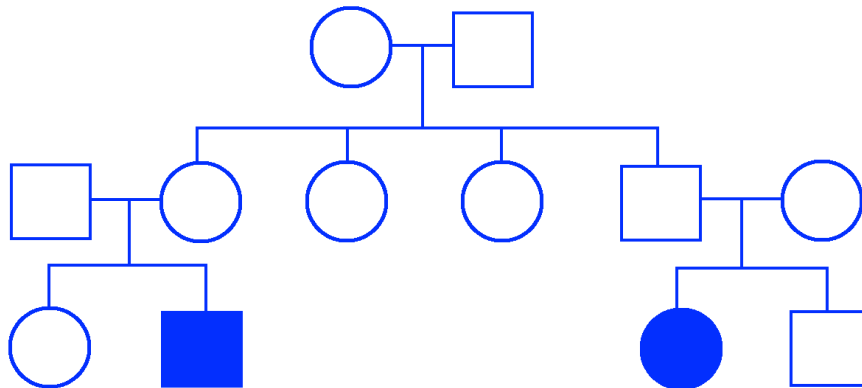
Karyotype analysis shows uniformity of karyotype among cells, indicating mosaicism is not the etiology. “Post-fertilization mitotic error” refers to somatic mosaicism, where a fraction of the patient’s cells would have trisomy 21.

A Robertsonian translocation is when two chromosomes become attached to each other, usually at their long arms. Such translocations can be *de novo* but are usually inherited by phenotypically normal parents with balanced translocations. In the case of Down syndrome, this may occur such that an additional copy of chromosome 21 is inherited due to its attachment to chromosome 14. The nomenclature is often written as [46,XX,t(14;21)] or [46,XY,t(14;21)].

WT1 gene mutations are associated with syndromes that manifest Wilms tumor. WAGR complex is one such syndrome that presents with aniridia, not Brushfield spots. WAGR is Wilms tumor, Aniridia, Genitourinary anomalies, Retardation.

Mutations of *WT2* classically cause Beckwith-Wiedemann syndrome, characterized by Wilms tumor, neonatal hypoglycemia, hemihypertrophy, macrosomia, and macroglossia.

18. A 6-year-old girl has had recurrent pneumonias since birth. Her mother reports that her stools appear to float and are foul-smelling. She is at the 5th percentile for height and weight. Chest auscultation reveals bilateral crackles. Chest x-ray shows increased bronchovascular markings. She is III-3 in the pedigree below.



Which of the following is the most likely explanation for the pattern of inheritance in this family?

- A) Autosomal dominant with incomplete penetrance
- B) Autosomal dominant with variable expressivity
- C) Autosomal recessive with high heterozygote frequency
- D) Autosomal recessive with low heterozygote frequency
- E) *De novo*

The correct answer is C.

The diagnosis is cystic fibrosis, an autosomal recessive disorder caused by mutations in the *CFTR* gene on chromosome 7, which result in a misfolded CFTR chloride channel.

In order for cousins III-2 and III-3 to have the disease despite the absence of disease in other family members, the implication is that the carrier (heterozygosity) rate is high.

The misfolded CFTR chloride channel remains sequestered within the rough endoplasmic reticulum (i.e., it is not transported to the cell surface).

CFTR channel is cAMP-mediated and ATP-gated.

Bottom line: Cystic fibrosis is autosomal recessive. Know this inheritance pattern for USMLE.

19. An 8-year-old boy is brought to the pediatrician by his parents for progressive muscle weakness. Physical examination shows scoliosis and enlargement of the calf muscles. When positioned on his stomach and asked to stand up, the boy appears to “walk” to a standing position using his arms. If the boy’s parents have more children, what is the probability they will have a child who is a carrier for this condition?
- A) 0
 - B) 1 in 1
 - C) 1 in 2
 - D) 1 in 4
 - E) 1 in 8
 - F) 2 in 3

The correct answer is D.

Duchenne muscular dystrophy is an X-linked recessive disorder usually caused by frameshift mutations in the *DMD* gene coding for dystrophin.

Dystrophin is a protein required for maintaining the integrity of the muscle cell by linking its cytoskeleton to the extracellular matrix.

Boys cannot be carriers for this condition because it is XR; only girls can be carriers.

The chance that the parents have another child who is a girl is 1/2. There is then a 1/2 chance that the mother, who is a carrier, passes the X-chromosome with the *DMD* mutation to the daughter.

Therefore, $1/2$ (chance of daughter) $\times$ $1/2$ (chance of passing mutated X) = 1 in 4 chance of having a child who is a carrier for this condition.

Physical examination in Duchenne frequently shows pseudohypertrophy of the calf muscles. The enlargement of the calves is not pure muscle; fibroadipose deposition ensues leading to weakness. Most boys lose the ability to walk by adolescence.

Gower's maneuver is the use of the arms to stand up from a prone position. This is in order to compensate for weakness of the pelvic muscles.

Becker muscular dystrophy is a less severe form of Duchenne that is also an XR disorder of the *DMD* gene. Unlike Duchenne, which is classically caused by frameshift mutations and presents within the first few years of life, Becker is frequently caused by missense mutations and presents in late adolescence or early adulthood.

20. A phenotypically normal 24-year-old woman is a carrier of an X-linked recessive disorder that is lethal in utero (i.e., results in spontaneous abortion). Which of the following is the probability that the next child she gives birth to is a carrier for this disorder?
- A) 1 in 1
 - B) 1 in 2
 - C) 1 in 3
 - D) 1 in 4
 - E) 2 in 3
 - F) 3 in 4

The correct answer is C.

Half of the males this patient conceives will be spontaneously aborted in utero, as there is a 1/2 chance a male child will receive the lethal X-chromosome.

In contrast, no females will be aborted in utero because this is an XR condition.

Our possible combinations for conception are:

- Normal female (non-carrier): XX
- Carrier female: XX
- Normal male: XY
- ~~Diseased male (lethal): XY~~

This means there is a 2:1 ratio of females:males who will be birthed by this patient, where 1:3 of the children overall will be carriers.

21. A 28-year-old male presents to the physician for a 3-day history of dark urine. Dipstick shows 2+ blood. He has had many such occurrences in the past. The patient wears hearing aids and glasses. Renal biopsy shows an apparent splitting of the glomerular basement membrane. Which of the following inheritance patterns best describes this condition?
- A) Autosomal dominant
 - B) Autosomal recessive
 - C) Mitochondrial
 - D) Polygenic
 - E) X-linked

The correct answer is E.

The diagnosis is Alport syndrome, an X-linked disorder caused by mutations in the gene for collagen type IV. The vignette will give you a male with hematuria who has an ear and/or eye problem.

Renal biopsy will show splitting of the basement membrane with a characteristic “basket-weave” appearance.

Patients can develop sensorineural hearing loss and diminished visual acuity due to abnormal collagen IV in the inner ear and lens of the eye, respectively. Hematuria results from loss of glomerular basement membrane integrity.

Do not confuse Alport with Goodpasture syndrome, which is instead caused by antibodies against type IV collagen.

Goodpasture will present as a male 20s-40s with hematuria and hemoptysis. Renal biopsy shows linear immunofluorescence due to anti-glomerular basement membrane (anti-GBM; type IV collagen) antibodies.

22. An 11-year-old boy is brought to the pediatrician by his mother for a follow-up examination. Physical examination shows everted ears and a long jaw. Testicular volume is increased. Karyotypic analysis shows an expansion of 650 CGG repeats within the 5' untranslated region of the *FMR1* gene. Which of the following is the most likely effect of this CGG expansion?
- A) Decreased transcription
 - B) Decreased translation
 - C) Hypermethylation causing increased gene activation
 - D) Impaired mRNA splicing
 - E) Impaired 5-methylguanosine capping of mRNA

The correct answer is A.

Fragile X syndrome is an X-linked dominant disorder caused by a CGG trinucleotide expansion within the *FMRI* gene. Greater than 200 CGG repeats results in disease.

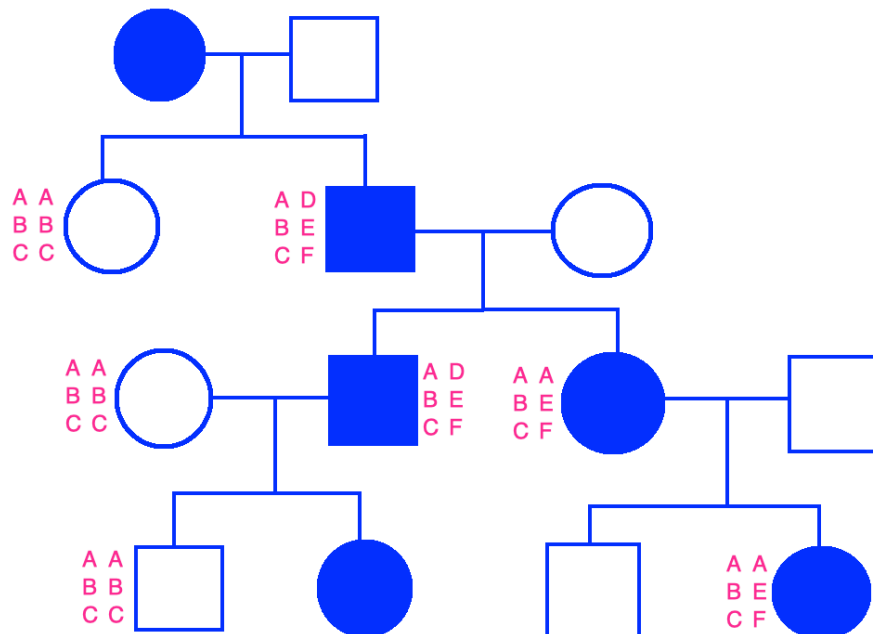
Prominent, everted ears, a long, narrow jaw, and reduced IQ are characteristic of the phenotype. Males will also often have macroorchidism.

The CGG repeat expansion results in hypermethylation of the *FMRI* gene and resultant decreased binding of transcription factors and RNA polymerase II. The answer on the USMLE is “decreased transcription.”

The high-yield trinucleotide repeat disorders for USMLE are:

- FraGile X: CGG → X-chromosome; XD.
- Huntingon disease: CAG → chromosome 4; AD.
- MyoTonic dystrophy: CTG → chromosome 19; AD.
- Friedreich Ataxia: GAA → chromosome 9; AR.

23. A family has many individuals with phenotypes characterized by hyperpigmented macules, axillary and groin freckling, iris hamartomas, pheochromocytoma, and optic glioma. Genetic analysis of chromosome 17 is performed. Patients III-3 and IV-4 in the pedigree bear disease phenotype but have an observed wild-type locus on the examined allele.



Which of the following is the most likely explanation for these findings?

- A) Insertion
- B) Inversion
- C) Recombination
- D) Single nucleotide polymorphism
- E) Transposition

The correct answer is C.

The diagnosis is neurofibromatosis type I (NF1), an autosomal dominant disorder on chromosome 17 demonstrating complete penetrance and variable expressivity.

The wild-type (non-diseased) allele in this family is identified as A-B-C. The individuals with NF1 have the diseased D-E-F allele.

However, patients III-3 and IV-4 are observed to have allele A-E-F, rather than D-E-F. This is because locus A from the wild-type allele replaced locus D on the diseased allele during meiosis-I, where chromatids exchange some genetic material and then separate. This is called recombination. The chance that two loci will undergo a recombination event is called the recombination fraction.

Single-nucleotide polymorphisms (SNPs) are when one base substitutes for another. These account for most variation among individuals. While an SNP could explain a random alteration in the diseased locus, it would not specifically account for the wild-type locus having become the one present on the diseased allele.

Insertions, inversions, and transpositions are known chromosomal rearrangements, but the replacement of a diseased locus with a wild-type one is due to recombination during meiosis-I.

An insertion might be observed as D-A-E-F instead of the D-E-F locus.

A transposition is when a piece of genetic material from one part of the genome migrates and inserts itself elsewhere within the genome. If a transposition of a wild-type locus occurred, it would be observed as an insertion event.

An inversion might be observed as D-F-E instead of D-E-F.

24. A 34-year-old woman presents to the physician for a lump she self-palpated in her right breast. Her mother was diagnosed with breast cancer at the age of 40; her maternal aunt was diagnosed with ovarian cancer at the age of 37. Genetic testing is most likely to show a mutation in which of the following molecular processes?
- A) Base excision repair
 - B) Mismatch repair
 - C) Nucleotide excision repair
 - D) Recombinational double-stranded DNA repair
 - E) Slipped-strand mispairing

The correct answer is D.

BRCA1 and *BRCA2* are tumor suppressor genes on chromosomes 17 and 13, respectively. Inheritance is autosomal dominant with incomplete penetrance (i.e., cancer phenotype can “skip” a generation).

Mutations result in increased risk of breast cancer in both females and males, as well as gynecologic cancers in females and testicular cancer in males.

Mutations in *BRCA* genes result in defective recombinational double-stranded DNA repair. This process normally involves the repair of DNA using complementary strands from homologous dsDNA.

Base excision repair involves the removal of a single mutated base. This is the answer on the USMLE if the stem mentions glycosylases (i.e., glycosylase enzymes are required for this process).

Nucleotide excision repair involves the removal of bulky DNA distortions, namely pyrimidine-pyrimidine dimers due to UV exposure. This is the answer on the USMLE for xeroderma pigmentosa.

Mismatch repair involves the removal of an incorrect base inserted during DNA synthesis. The template strand is methylated; the newly synthesized strand is unmethylated. Mismatch repair enzymes identify the newly synthesized strand due to its lack of methylation and remove/replace the incorrect base.

Mismatch repair is the answer on USMLE for Lynch syndrome (HNPCC). This presents as colonic polyps +/- gynecologic cancers or other organ system cancers, e.g., pancreatic or duodenal cancer. Defects in mismatch repair are sometimes referred to as “microsatellite instability.” The important genes are *MLH1*, *MSH2*, *MSH6* and *PMS2*.

Slipped-strand mispairing is the answer on USMLE if there is an insertion or deletion of bases within strands of tandem repeats. For example, if there is a sequence of CCCCCCTA that erroneously becomes CCCCCCCTA, where a C was inserted, the type of repair that is needed here is slipped-strand mispairing.

25. An investigator is looking at a mutation sequence within the collagen type I gene of a patient with osteogenesis imperfecta. This sequence is compared to that of a healthy individual:

Healthy individual: 5'-GAGCCCTA-3'

Osteogenesis imperfecta: 5'-GAGGCCTA-3'

This mutation allows for the creation of a diagnostic test using a restriction enzyme. Which of the following sequences most likely represents the cleavage site for the restriction enzyme?

- A) 5'-ACCG-3'
- B) 5'-CAGC-3'
- C) 5'-CAGG-3'
- D) 5'-AGGC-3'
- E) 5'-GGCC-3'
- F) 5'-GCCA-3'

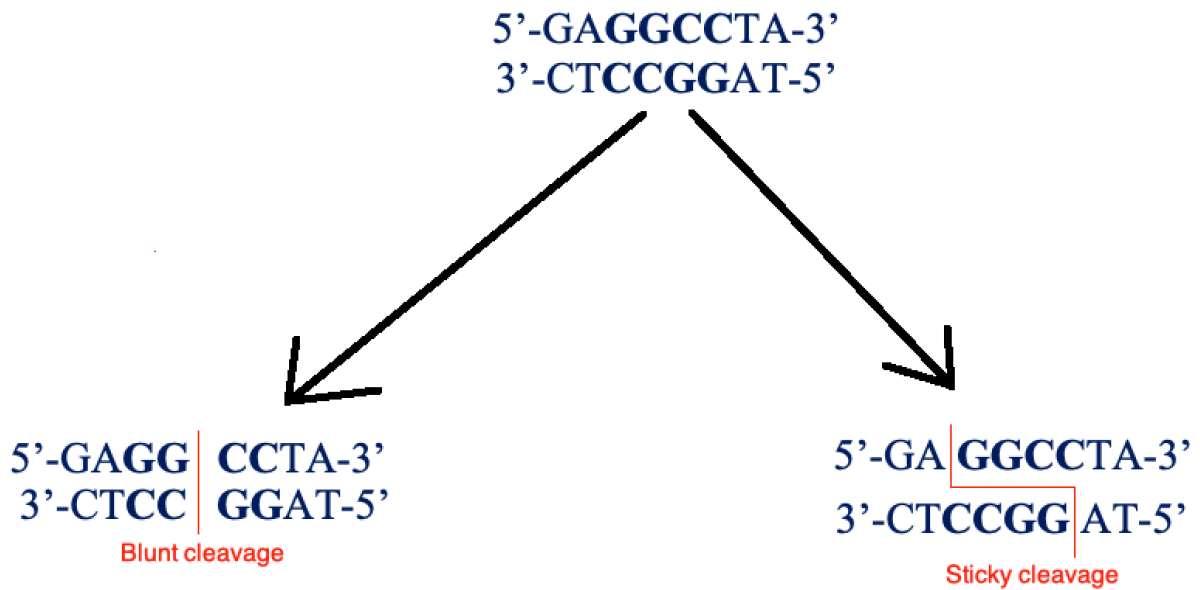
The correct answer is E.

Student says, “Wtf?”

The USMLE wants you to know that restriction enzymes usually cleave along palindromic sequences.

A palindrome is a sequence is one that is the same forward and backwards.

5'-GGCC-3' becomes palindromic when paired with 3'-CCGG-5' as dsDNA. The restriction enzyme will recognize this sequence and then cleave:



Bottom line: if you get an obscure restriction enzyme question and are asked to identify a cleavage site, remember that palindromic sequences are preferred by these enzymes.

26. The frequency of an autosomal recessive disease in the population is $1/40,000$. Frameshift mutations account for 30% of mutations at this locus; 70% are due to point mutations. The frequency of frameshift mutation carriers in the population is closest to which of the following?
- A) $2/125$
 - B) $3/100$
 - C) $3/1000$
 - D) $7/100$
 - E) $9/400$
 - F) $15/200$

The correct answer is C.

Using Hardy-Weinberg: $1 = p^2 + 2pq + q^2 = p + q$, $q^2 = 1/40,000$. Therefore $q = 1/200$.

Since $p + q = 1$, $p = 199/200 \approx 1$.

$2pq = 2 \times 1 \times 1/200 = 1/100 =$ carrier frequency in population.

However, of the $1/100$ in the population who are carriers, only 30% are frameshift mutation carriers. Therefore: $1/100 \times 3/10 = 3/1000$.

If the question asked for point mutation carriers, we'd have: $1/100 \times 7/10 = 7/1000$.

If that sounds confusing, think of it this way: since the carrier frequency in the population is $1/100$, that's the same as $10/1000$. So 30% of $10/1000 = 3/1000$ for frameshift mutation carriers, and 70% of $10/1000 = 7/1000$ for point mutation carriers.

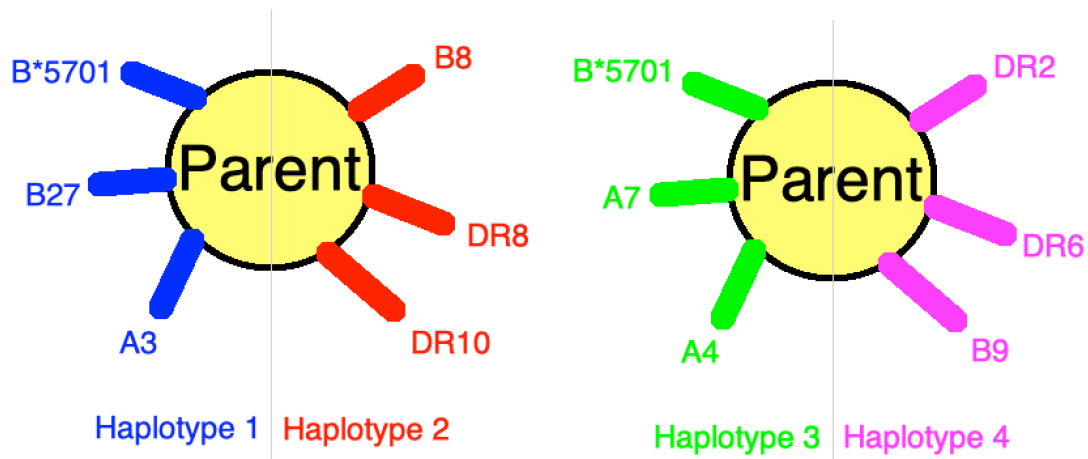
27. A 24-year-old woman needs to undergo a renal transplant secondary to long-standing history of granulomatosis with polyangiitis. She has a brother and sister. What is the probability that both her brother and sister will have human leukocyte antigen types that make them suitable donors?
- A) 1 in 1
 - B) 1 in 2
 - C) 1 in 4
 - D) 1 in 6
 - E) 1 in 8
 - F) 1 in 16
 - G) 1 in 32
 - H) 1 in 64
 - I) 1 in 128

The correct answer is F.

As discussed earlier, HLA genes are inherited as haplotypes from each parent. There is a 1/4 chance that an individual's full sibling will have the same HLAs (i.e., 1/2 chance of inheriting the same haplotype from mom x 1/2 chance of inheriting the same haplotype from dad).

This question asks for the probability that *both* siblings are suitable donors.

Therefore $1/4 \times 1/4 = 1/16$.



Possible offspring combinations

- Haplotype 1 Haplotype 3
- Haplotype 1 Haplotype 4
- Haplotype 2 Haplotype 3
- Haplotype 2 Haplotype 4

28. A 16-year-old girl is brought to the GP by her mom because she has not yet had a menstrual period. Physical examination shows a vagina that is 3-cm long and ends in a blind pouch. Her breasts are Tanner stage 4 and there is coarse pubic and axillary hair. Which of the following is the most likely karyotype of this patient?
- A) 45X
 - B) 46XX
 - C) 46XY
 - D) 47XXX
 - E) 47XXY

The correct answer is B.

The diagnosis is Müllerian agenesis, which results in atresia of the vagina, cervix, uterus, and Fallopian tubes, as these structures are derived from the Müllerian ducts.

The ovaries are not derived from the Müllerian ducts and thus are not affected. Normal secretion of hormones results in phenotypic female with normal Tanner stages.

If the USMLE gives you this same exact presentation but tells you there is scanty or absent pubic and axillary hair, the answer is androgen insensitivity syndrome, not Müllerian agenesis. This is due to a defect in the androgen receptor, and dihydrotestosterone (DHT) is needed for the development of external male sex characteristics. The patient will be a karyotypic male (46XY) but appear phenotypically female. In other words:

Female teenager + normal Tanner stages + vagina ends in blind pouch + normal/coarse pubic/axillary hair: answer = Müllerian agenesis. Karyotype = 46XX (karyotypic female).

Female teenager + normal Tanner stages + vagina ends in blind pouch + scanty/absent pubic/axillary hair: answer = androgen insensitivity syndrome. Karyotype = 46XY (karyotypic male).

Turner syndrome (45X) usually presents as a short female (<5'1") with a shield chest that is Tanner stage 1 or 2, cystic hygroma (webbed neck due to lymphatic insufficiency), and scattered nevi. Coarctation of the aorta and/or bicuspid aortic valve (aortic stenosis) may be seen. Patients with Turner present with streak ovaries and will not menstruate; this is colloquially referred to as "menopause before menarche." USMLE wants you to know pregnancy is possible via IVF with donor oocytes (i.e., for behavioral questions, it is incorrect to outright say she cannot have a pregnancy).

Klinefelter syndrome (47XXY) most often presents as a tall male with eunuchoid body shape (i.e., wide hips like a female) and small phallus and testes. Males cannot reproduce due to agenesis of the seminiferous tubules.

5 α -reductase deficiency (46XY) is another common vignette. The enzyme normally converts testosterone $\rightarrow$ DHT. It frequently presents as "phallus at age 12," where a 12- or 13-year-old girl grows four inches in the past four months and whose clitoris has elongated to 3-4 cm in length. This is due to a surge of testosterone during puberty that may overcome some of the enzyme deficiency, resulting in sufficient DHT for nascent external male sex characteristics. Another presentation will be a neonate born with ambiguous genitalia who is confirmed 46XY and who has male, but not female, genital ducts.

29. A 32-year-old woman presents to the physician for a one-week history of sore hands and knees. Physical examination shows an ulcer on the buccal mucosa. Laboratory studies show:

Hemoglobin	10.5 g/dL
Hematocrit	31.5%
WBC	4,500/ μ L
Platelets	60,000/ μ L

Which of the following is the most likely mode of inheritance of this condition?

- A) Autosomal dominant
- B) Autosomal recessive
- C) Mitochondrial
- D) X-linked dominant
- E) Multifactorial

The correct answer is E.

The diagnosis is systemic lupus erythematosus (SLE). USMLE vignettes will often omit descriptors such as malar (butterfly) rash because they're too easy.

Arthritis is the most common presenting feature of SLE (>90%).

Students should be aware of mucositis (mouth ulcers) and antibodies against hematologic cell lines as very high-yield in SLE. Antibodies are most frequently against platelets, resulting in isolated thrombocytopenia, but leuko- and erythropenia can also be seen. If there is an isolated erythropenia (i.e., no thrombocytopenia), then anemia of chronic disease should be suspected instead.

Most autoimmune diseases (i.e., sarcoidosis, thyroiditis, pernicious anemia, etc.) do not carry a strict mode of inheritance and are best characterized as multifactorial – i.e., associated with many different gene and HLA combinations. This also applies to diabetes mellitus type I, androgenetic alopecia, pyloric stenosis, and cleft/lip palate.

30. A couple gives birth to a child with a cleft lip. Which of the following is the most likely probability they will have a second child with a cleft lip or palate?
- A) 0%
 - B) 2-4%
 - C) 10%
 - D) 25%
 - E) 50%
 - F) Same as general population since clefts are sporadic and multifactorial

The correct answer is B.

If a couple gives birth to a child with a cleft lip or palate, and neither parent has a cleft, the chance of a subsequent resulting in cleft lip or palate is approximately 2-4%.

If one of the parents has a cleft lip or palate, the chance the child will have a cleft is closer to 5%.

This same probability applies to congenital heart defects (i.e., ASD, VSD, etc.). The chance of recurrence is 2-4% (on NBME).

31. A 23-year-old man is recently diagnosed with schizophrenia. He has an identical twin brother. What is the probability his twin brother will develop schizophrenia?
- A) 5%
 - B) 10%
 - C) 25%
 - D) 50%
 - E) 75%

The correct answer is D.

The USMLE asks this for some reason. You need to know that if an individual develops schizophrenia, the chance of his or her monozygotic (identical) twin also developing schizophrenia is approximately 50%.

Factoidy? Agreed. But it's not our opinion that matters. It's what shows up on the exam that does.

32. A two-week-old male vomits forcefully after feeds. The vomitus is non-bilious. Diagnosis is confirmed using abdominal ultrasound. Which of the following best describes this condition?
- A) Autosomal dominant; part of syndromic disorder
 - B) Autosomal recessive; part of syndromic disorder
 - C) Autosomal recessive; one-off developmental anomaly
 - D) Multifactorial; one-off developmental anomaly
 - E) X-linked; seen most often in first-born males

The correct answer is D.

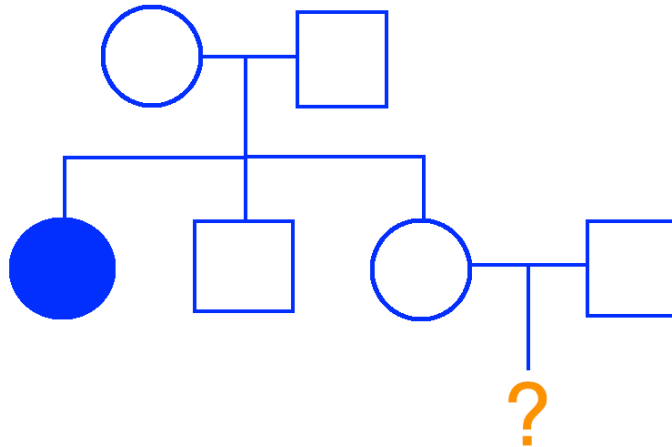
The highest yield genetics point about pyloric stenosis is that it is a one-off developmental anomaly – i.e., it is not part of a greater syndrome. This is in contrast to duodenal atresia and Hirschsprung, which are frequently associated with Down syndrome. This is asked on the NBMEs for Step 1.

Pyloric stenosis is considered to be most often a multifactorial anomaly with no strict inheritance. Rarely, it may be considered to be autosomal dominant in some pedigrees, however this is not the primary mode of inheritance.

Pyloric stenosis is best considered as part of a multifactorial inheritance. As discussed earlier, this is similar to cleft lip/palate and androgenetic alopecia, as well as autoimmune diseases such as diabetes mellitus type I, sarcoidosis, and SLE.

Pyloric stenosis demonstrates a greater prevalence in first-born males, however it is not X-linked.

33. A 24-year-old woman (II-1 in the pedigree) has an autosomal recessive disorder. The frequency of the condition in the population is $1/40,000$.



What is the probability that II-3 and II-4 will have a son who is a carrier for this condition?

- A) $1/200$
- B) $1/250$
- C) $1/300$
- D) $1/350$
- E) $1/600$
- F) $1/800$

The correct answer is E.

Using Hardy-Weinberg: $1 = p^2 + 2pq + q^2 = p + q$

$q^2 = 1/40,000$, therefore $q = 1/200$ and $p = 199/200 \approx 1$.

Carrier frequency = $2pq = 2 \times 1 \times 1/200 = 1/100$ = chance II-4 is a carrier.

Since II-3 is a phenotypically normal sibling of a patient (II-1) with an autosomal recessive disease, the chance II-3 is a carrier is $2/3$.

Therefore, the chance II-3 and II-4 have a child who is a carrier for the disorder is:

$2/3 \times 1/100 \times 1/2$ (since the Punnett square will have $2 \times Aa$) = $2/600 = 1/300$.

However, the question asks for the probability the couple will have a *son* who is a carrier, so we need to multiply this by the chance they have a son: $1/300 \times 1/2 = 1/600$.

34. A healthy 25-year-old man has two siblings with cystic fibrosis. The carrier frequency in the general population is $1/25$. Assuming no *de novo* mutations, what is the probability he will have a daughter who does not carry a cystic fibrosis mutation?
- A) $1/50$
 - B) $1/100$
 - C) $1/150$
 - D) $1/200$
 - E) $1/300$
 - F) $1/900$

The correct answer is E.

The patient has siblings with cystic fibrosis. This means there is a $2/3$ chance he is a carrier (i.e., he is healthy, so we can eliminate “aa” from his Punnett square, leaving 2x Aa and 1x AA as his possibilities).

The carrier frequency in the general population is $1/25$.

So $2/3 \times 1/25 \times 1/4$ (chance of having a child with AA in Punnett square; no disease allele) = $2/300 = 1/150$.

However, the question asks for the chance he will have a *daughter* who does not have a diseased allele, so we need to multiply by the chance he will have a daughter ($1/2$).

So $1/150 \times 1/2 = 1/300$.

35. A 20-year-old woman and her 47-year-old husband come to the physician for preconception genetics counseling. Both patients have increased HbA₂ on hemoglobin electrophoresis. Genetic analysis shows the husband has heterozygosity for a null mutation in the β -globin gene. The woman is heterozygous for an allele that results in a 50% reduction in β -globin synthesis. Which of the following best represents the β -globin possibilities for this couple's offspring?
- A) 1 in 2 will have normal β -globin function
 - B) 1 in 2 will have 25% β -globin function
 - C) 1 in 2 will have 50% β -globin function
 - D) 1 in 4 will have 75% β -globin function
 - E) 1 in 4 will carry a null allele

The correct answer is D.

		Female patient	
		β	$1/2\beta$
Male patient	β	$\beta\beta$	$\beta + 1/2\beta = 3/2\beta$
	$\emptyset$	$\beta + \emptyset = \beta$	$1/2\beta + \emptyset = 1/2\beta$

$\beta\beta = 100\% \text{ function}$
 $3/2\beta = 75\% \text{ function}$
 $\beta = 50\% \text{ function}$
 $1/2\beta = 25\% \text{ function}$

1 in 4 children will have 100%, 75%, 50%, and 25% β -globin function, respectively.

Choice A is wrong because 1 in 4, not 1 in 2, will have normal β -globin function.

Choice B is wrong because 1 in 4, not 1 in 2, will have 25% β -globin function.

Choice C is wrong because 1 in 4, not 1 in 2, will have 50% β -globin function.

Choice E is wrong because 1 in 2, not 1 in 4, will carry the null allele.

36. In a clinical study of a disease, 225 people are examined for single nucleotide polymorphisms. The results are shown below.

Genotype	Number of people
DD	60
DF	90
FF	75

If D is one allele and F is the other, which of the following is the frequency of allele D?

- A) $1/5$
- B) $1/9$
- C) $4/15$
- D) $7/15$
- E) $24/45$
- F) $150/225$
- G) $165/225$

The correct answer is D.

This question is “hard.”

If each of the 225 people in the study have two alleles, then there are 450 total alleles.

Solving for frequency of D allele:

$$60 \text{ people} \times (2D) = 120D \text{ alleles}$$

$$90 \text{ people} \times (1D) = 90D \text{ alleles}$$

$$120D + 90D = 210D$$

$$210D / 450 \text{ total alleles} = 70/150 = 7/15$$

Solving for frequency of F allele:

$$75 \text{ people} \times (2F) = 150F$$

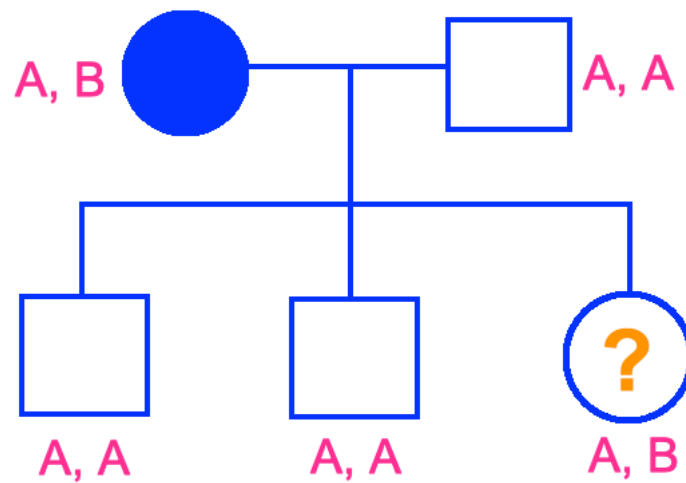
$$90 \text{ people} \times (1F) = 90F$$

$$150F + 90F = 240F$$

$$240F / 450 \text{ total alleles} = 240/450 = 24/45$$

$D = 210/450$; $F = 240/450$, where $210/450 + 240/450 = 450/450$ alleles total.

37. A 29-year-old woman at 20 weeks' gestation has an autosomal dominant disorder with 60% penetrance. A single nucleotide polymorphism (SNP) marker is identified on the diseased allele with a recombination fraction of 20%. Alleles A and B for this SNP marker are indicated on the pedigree.



Which of the following is the probability the fetus will demonstrate disease phenotype?

- A) 12%
- B) 18%
- C) 36%
- D) 38%
- E) 48%
- F) 62%
- G) 72%
- H) 84%
- I) 90%

The correct answer is E.

The B marker in and of itself does not result in disease. It is localized geographically nearby to the disease mutation on the chromosome.

Recombination occurs during prophase of meiosis-I, where chromatids crossover and exchange small amounts of genetic material, resulting in the unlinking of loci on the same chromatid.

Since there is a recombination fraction of 20%, there is a 20% chance the fetus received the B marker because it unlinked from the disease allele. This means there's an 80% chance she still has the B marker linked to the disease allele, and thus carries the disease allele.

Since this condition only bears 60% penetrance, the chance the fetus will demonstrate the disease phenotype is:

$$8/10 \times 6/10 = 48/100 = 48\%.$$

38. A 4-year-old girl has short stature. Both of her parents are of average stature. Genetic analysis shows paternal uniparental disomy of a particular locus. Which of the following mechanisms might explain the short stature?
- A) Maternal imprinting of one allele
 - B) Maternal imprinting of two alleles
 - C) Paternal imprinting of one allele
 - D) Paternal imprinting of two alleles

The correct answer is D.

Maternal imprinting = gene coming from mom is silenced; requires receiving normal allele from dad in order to not have disease.

Paternal imprinting = gene coming from dad is silenced; requires receiving normal allele from mom in order to not have disease.

The girl has received both of her alleles from dad (i.e., paternal uniparental disomy) and has disease.

This means both alleles are silenced coming from dad, which is the equivalent of having no allele at all.

This concept is similar to Angelman and Prader-Willi syndrome, as discussed earlier.

Angelman syndrome = paternal imprinting + deleted/mutated allele from mom, OR paternal uniparental disomy (where both alleles can by all means be healthy but are silenced).

Prader-Willi syndrome = maternal imprinting + deleted/mutated allele from dad, OR maternal uniparental disomy (where both alleles can by all means be healthy but are silenced).

39. A 19-year-old man presents to the physician with abdominal pain, pink urine, and decreased sensation in his left hand. His symptoms are mitigated with administration of hematin. Laboratory studies show deficiency of porphobilinogen deaminase. Which of the following inheritance patterns best describes this condition?
- A) Autosomal dominant
 - B) Autosomal recessive
 - C) Mitochondrial
 - D) X-linked dominant
 - E) X-linked recessive

The correct answer is A.

Acute intermittent porphyria (AIP) is an autosomal dominant disorder of the heme synthesis pathway. It is caused by deficiency of porphobilinogen deaminase. The classic presentation is abdominal pain, red urine, and neuropathy. Symptomatology is improved with administration of hematin first-line; glucose may also be effective.

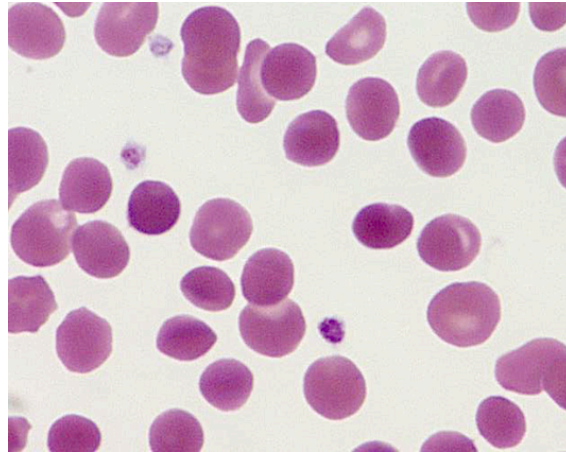
Inheritance of AIP is asked on an offline NBME. Questions are known to repeat on the real USMLE, so this is worth knowing.

Porphyria cutanea tarda, another heme synthesis disorder, is 20% of the time autosomal dominant and 80% of the time sporadic. This is due to deficiency of uroporphyrinogen decarboxylase. Disease presentation is classically photosensitivity (blistering) and tea-colored urine.

40. A 10-year-old boy is brought to the physician because of a 1-week history of fatigue. He recently recovered from a cold. Physical examination shows scleral icterus. Laboratory studies show:

Hemoglobin:	10.4 g/dL
Hematocrit:	31.2%
MCV:	92 fL
Direct antiglobulin test:	Positive

A blood smear is performed and shows spherocytes.



Genetic testing is most likely to show which of the following?

- A) Heterozygosity in the spectrin gene
- B) Heterozygosity in the α -globin gene
- C) Heterozygosity in the β -globin gene
- D) Homozygosity in the spectrin gene
- E) Homozygosity in the α -globin gene
- F) Homozygosity in the β -globin gene
- G) No specific mutation

The correct answer is G.

Students freak out every time they see this question and think there's an erratum.

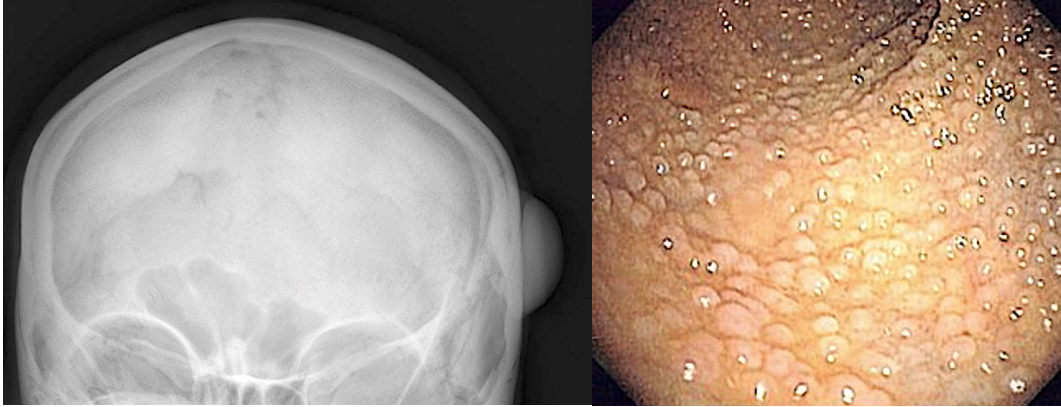
Spherocytes can be seen in drug- and infection-induced hemolysis. They are not limited to hereditary spherocytosis. This is high-yield for USMLE.

Hereditary spherocytosis is an autosomal dominant disorder most often in the ankyrin, spectrin, or band protein genes. Since it is AD, almost all individuals will only have one mutation (choice A).

However, this patient has a positive antiglobulin (Coombs) test. This indicates there are antibodies against RBCs as the etiology for the hemolysis. In hereditary spherocytosis, there is a negative Coombs test because the disorder has nothing to do with antibodies; there is merely an RBC cytoskeletal defect due to mutation in ankyrin, spectrin, or band protein genes.

The positivity of the Coombs test indicates that a drug or infection has caused the hemolysis. The child recently had an upper respiratory tract infection. Viral etiology is a known precipitator of hemolysis, e.g., CMV.

41. A 27-year-old man presents to the physician because of a 3-month history of a bump on his skull. His father died of colon cancer at the age of 36. A radiograph of the patient's skull and his colonoscopy are shown.



Which of the following best describes this patient's presentation?

- A) Gardner syndrome; autosomal dominant
- B) Gardner syndrome; autosomal recessive
- C) Lynch syndrome; autosomal dominant
- D) Lynch syndrome; autosomal recessive
- E) Peutz-Jeghers syndrome; autosomal dominant
- F) Peutz-Jeghers syndrome; autosomal recessive
- G) Turcot syndrome; autosomal dominant
- H) Turcot syndrome; autosomal recessive

The correct answer is A.

The patient has familial adenomatous polyposis (FAP), an autosomal dominant disorder caused by mutations in the adenomatous polyposis coli (*APC* gene) on chromosome 5.

Gardner syndrome = FAP + soft tissue tumors (e.g., lipomas) or bone tumors (e.g., osteomas). The image shown is an osteoma of the skull. It is not a brain tumor.

Turcot syndrome = FAP + CNS tumors. The image shown is an osteoma of the skull. It is not a brain tumor.

There is a 100% chance of development of colorectal cancer in FAP. Treatment is total proctocolectomy at age 18 (2CK surgery shelf Q), not repeated colonoscopies.

If the USMLE asks for the chance of this patient's child going on to develop colon cancer, the answer is 50% – i.e., 50% chance of passing along the mutated *APC* gene x 100% chance of cancer.

FAP is characterized by hundreds to thousands of polyps observed on colonoscopy. In other words, there are “lots of polyps.” In contrast, Lynch syndrome (HNPCC) presents with “polyps but not as many as FAP.” FAP is notably described as innumerable polyps, as seen in the image.

Lynch syndrome is autosomal dominant. It is caused by mutations in mismatch repair genes (*MSH2*, *MSH6*, *MLH1*, *PMS2*), resulting in microsatellite instability. Patients may also develop gynecologic cancers, or cancers of other organ systems (e.g., pancreas, duodenum).

Peutz-Jegher is autosomal dominant and characterized by perioral melanosis (hyperpigmentation on the lips and around the mouth) and hamartomatous polyps. USMLE will show you a picture of hyperpigmentation of the lips and then the answer will just be “hamartomatous” for the polyp type.

42. A couple has three children. Two of them have achondroplasia. Both parents are phenotypically normal. Which of the following best explains the disease pattern in this family?
- A) Germline mosaicism
 - B) Heteroplasmy
 - C) Imprinting
 - D) Incomplete penetrance
 - E) Variable expressivity

The correct answer is A.

This is a favorite USMLE question.

Achondroplasia is an autosomal dominant disorder caused by mutations in the fibroblast growth factor receptor gene, *FGFR3*. Most cases of achondroplasia occur, however, when both parents are phenotypically normal. This is because of germline (gonadal) mosaicism.

Germline (gonadal) mosaicism is when the spermatogonia or oogonia of one of the parents carry a mutation, but the somatic (body) cells do not. The result is that offspring can have an AD disorder despite the parents not being affected. Germline mosaicism is the answer classically for achondroplasia – i.e., the parents are normal but the child has the disease. The child will have 100% of his or her body cells affected because the original mutation was in the spermatogonia or oogonia of the parent, rather than occurring post-mitotically, as with somatic mosaicism. If the USMLE does not give achondroplasia, they will say two phenotypically normal parents have two children with an AD disorder, implying there is a germline mutation in one of the parents (usually the father).

Incomplete penetrance is when not everyone with the diseased genotype bears the diseased phenotype (i.e., not everyone with the disease expresses it); this is sometimes referred to as “skipping a generation,” and also applies to the *BRCA* genes.

Variable expressivity, in contrast, implies 100% penetrance – i.e., everyone with the disease expresses it in some form. The difference is the mere severity in which family members manifest the condition. *NF1* is a classic example, where everyone with the disease has it, but whilst one family member might only present with axillary and groin freckling, another might develop pheochromocytoma and optic glioma. **If the NBME/USMLE gives you a pedigree that looks like incomplete penetrance, but the diagnosis in the family is *NF1*, choose variable expressivity, not incomplete penetrance.** A patient with minimal disease severity in *NF1* might not report having it.

Heteroplasmy is a phenomenon seen in mitochondrial disorders, an inheritance pattern characterized by strict maternal transmission (i.e., mothers transmit disease to sons and daughters; fathers never transmit disease), where varying amounts of diseased mitochondria are passed to the next generation, resulting in offspring with disease of varying severity. This is due to the mother’s oocytes carrying thousands of copies of mitochondrial DNA, where the oocytes differ in the ratios of diseased mitochondria they contain. Do not confuse this with variable expressivity, which instead refers to non-mitochondrial disorders, where a chromosomal mutation can result in varying severity of disease.

Imprinting is when a gene is silenced coming from one parent. As mentioned earlier:

Angelman syndrome = paternal imprinting + deleted/mutated allele from mom, OR paternal uniparental disomy (where both alleles can by all means be healthy but are silenced).

Prader-Willi syndrome = maternal imprinting + deleted/mutated allele from dad, OR maternal uniparental disomy (where both alleles can by all means be healthy but are silenced).

43. A 16-year-old boy has two exons of his β -globin gene shown.

```

1      gggtagatcg gcctatatac ttatatattg atctagatcg aaacgaatcg
51     atctagatcg ttttatattg gcgtacatcg attCGAATCG ACCTGTATAG
101    TGAGCGGATC CGatatattg Gcctatatac ataccgatcg ACTGGCGATA
151    CGGTGTTTAC ccgtacattg aattacattc gcctttatcg atccagaccc

```

Exons are given in bold capital letters. A mutation C→G at position 121 (red capital) is most likely to result in β -thalassemia due to which of the following mechanisms?

- A) Disruption of post-transcriptional processing
- B) Impairment of post-translational modifications
- B) Missense mutation resulting in misfolded protein
- C) Nonsense mutation resulting in non-functional, truncated protein

The correct answer is A.

Letters that are bold and capitalized are exons. The patient therefore has a mutation within one of his intron sequences.

Mutations within intron sequences are frequently splice-site mutations, resulting in impaired post-transcriptional processing. The USMLE might describe this type of mutation as “impairment of normal splicing by creation of new splice site.”

Transcription = DNA → RNA

Translation = RNA → protein

When DNA is transcribed into RNA within the nucleus, the nascent RNA is known as heteronuclear RNA (hnRNA). It then undergoes three modifications within the nucleus:

- 1) Addition of 3' poly-A tail
- 2) Addition of 5' methylguanosine cap
- 3) Splicing out of introns

After these three modifications occur within the nucleus, the hnRNA is now called mRNA. The mRNA then leaves the nucleus for the cytosol, where it is translated into protein. mRNA can also dock with P-bodies in the cytosol. These are storage sites for mRNA to be used later, or where additional modifications may occur.

The splicing process occurs post-transcription, not post-translation.

Do not confuse this process with the mechanism for Fragile X. As discussed earlier, an expansion of CGG >200 repeats within the 5'-untranslated region of the *FMR1* gene results in hypermethylation and decreased transcription of the gene, *not* impaired splicing.

44. A 46-year-old man presents to the physician because of a painful big toe. Allopurinol has been ineffective in reducing recurrences.



Which of the following is the most likely inheritance pattern for this condition?

- A) Autosomal dominant; complete penetrance; not syndromic
- B) Autosomal dominant; incomplete penetrance; not syndromic
- C) Autosomal recessive; not syndromic
- D) X-linked recessive; Lesch-Nyhan syndrome
- E) No specific enzyme deficiency or heritability

The correct answer is E.

The image shows podagra (gout of the first metatarsophalangeal joint).

Some gene loci have been identified that have been posited to increase the risk of episodes, but gout is almost always an idiopathic condition not associated with any specific type of enzyme deficiency or inheritance pattern.

High-yield risk factors include alcoholism and consumption of purine-rich foods such as anchovies and organ meats.

Lesch-Nyhan syndrome is an XR condition leading to uric acid overproduction and gout. However this is a pediatric condition. The vignette will be a male infant who exhibits self-mutilation and has red-orange sandy urate crystals in his diaper.

45. A 23-year-old man presents to the physician because of a yellowish 1-cm growth over his left Achilles tendon. Laboratory studies show increased serum triglycerides and a milky appearance of the plasma following centrifugation. Enzyme analysis shows decreased activity of lipoprotein lipase. Which of the following is the most likely inheritance pattern of this condition?
- A) Autosomal dominant
 - B) Autosomal recessive
 - C) X-linked dominant
 - D) X-linked recessive
 - E) Sporadic

The correct answer is B.

The diagnosis is familial hyperchylomicronemia, an autosomal recessive condition caused by deficiency of lipoprotein lipase or apolipoprotein C-II. Serum triglycerides and chylomicrons will be elevated. The serum bears a characteristic milky or creamy appearance upon centrifugation.

In contrast, familial hypercholesterolemia and familial hypertriglyceridemia are both autosomal dominant.

Familial hypercholesterolemia causes an isolated increase in serum LDL. Heterozygosity will yield total cholesterol usually in the ~3-400s mg/dL. Homozygosity will yield ~700-1000 mg/dL. The mechanism is deficiency of LDL receptor (heterozygosity), or complete absence (homozygosity).

Familial hypertriglyceridemia causes elevated serum triglycerides. The plasma is not described as milky or creamy upon centrifugation. The mechanism is increased hepatic production of VLDL.

46. A 25-year-old man presents to the physician because of a two-week history of abdominal pain after eating meals. He is prescribed omeprazole. Two years ago he was briefly hospitalized for left-sided shooting groin pain and underwent a lithotripsy procedure. Which of the following is the most likely probability one of his daughters will also have this condition?
- A) 0%
 - B) 25%
 - C) 50%
 - D) 75%
 - E) 100%
 - F) Cannot be determined without carrier status of partner

The correct answer is C.

The diagnosis is multiple endocrine neoplasia type I (MEN 1). The MEN syndromes are inherited in an autosomal dominant fashion.

The patient has Zollinger-Ellison syndrome (gastrinoma) resulting in duodenal ulcers.

The history of ureterolithiasis is secondary to hypercalcemia due to primary hyperparathyroidism.

MEN 1: Pancreas, pituitary, parathyroid adenoma or hyperplasia
 MEN1 gene; chromosome 11

MEN 2A: Medullary thyroid carcinoma, pheochromocytoma, parathyroid adenoma or hyperplasia
 RET GENE; chromosome 10

MEN 2B: Medullary thyroid carcinoma, pheochromocytoma, Marfanoid body habitus, mucosal
 neuromas
 RET GENE; chromosome 10

Since MEN 1 is autosomal dominant, there is a 50% chance one of his daughters will have the condition.

The question does not ask for the probability he will have a daughter with the condition (25%).

47. A researcher wishes to investigate clonality of a population of leukemic cells. Analysis of which of the following would be best to determine monoclonality?
- A) IL-2 receptor subunits
 - B) IL-4 receptor subunits
 - C) Integrin density
 - D) Mitotic rate
 - E) X-chromosome-linked isoenzymes

The correct answer is E.

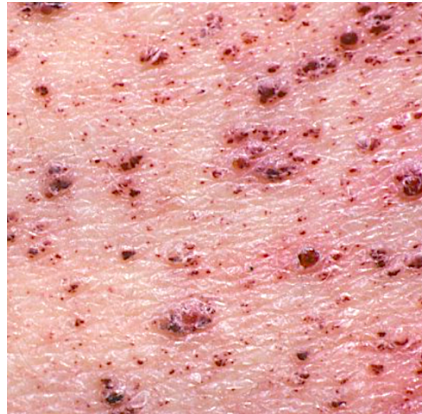
Yes, this is a wtf question. But it's not our opinion that matters. It's what shows up on the USMLE that does.

Clonality refers to the source of certain cell types. For instance, most cancers are monoclonal, where they can ultimately be traced back to one cell that became neoplastic and then replicated. Polyclonal means a cell population is derived from many different cells.

The most commonly used method for determining clonality is based on detection of X-chromosome sequence polymorphisms via identification of protein isoforms, methylation status, and transcribed mRNA.

Bottom line: if you get a weird question about clonality, choose "X-chromosome-linked isoenzymes."

48. A 27-year-old man presents to the physician for a follow-up appointment. He has chronic kidney disease and restrictive cardiomyopathy. Previous enzyme analysis demonstrated a deficiency of α -galactosidase A. The patient's torso is shown in the photograph below.



If the carrier frequency is 1/100,000, which of the following is the disease frequency in the population?

- A) 1 in 25,000
- B) 1 in 50,000
- C) 1 in 100,000
- D) 1 in 200,000
- E) 1 in 345,000
- F) 1 in 480,000

The correct answer is D.

Fabry disease is an XR lysosomal storage disease due to deficiency of α -galactosidase A and accumulation of ceramide trihexoside.

The classic presentation is a male with angiokeratomas (benign cutaneous capillary growths in the photograph) who has renal and/or cardiac pathology.

This question is the reverse of Q16, where we solve for carrier frequency using disease frequency as a starting point. In question, we start with carrier frequency and solve for disease frequency.

For XR disorders, disease frequency in males = q ; disease frequency in females = q^2 .

Since males cannot be carriers for XR disorders, carrier frequency in females = carrier frequency in the population.

Using Hardy-Weinberg, carrier frequency = $2pq$, where $p + q = 1$.

Since $2pq = 1/100,000$, and $p \approx 1$, we have $2 \times 1 \times q = 1/100,000 = 2q$; therefore $q = 1/200,000$.

Had the question instead given us $1/200,000$ as the disease frequency to start with, we would say:

$2pq = 2 \times 1 \times 1/200,000 = 1/100,000 = \text{carrier frequency}$.

It should be noted that of the lysosomal storage diseases, Fabry and Hunter are both XR.

In contrast, Hurler, Gaucher, Tay-Sachs, Niemann-Pick, Krabbe, and metachromatic leukodystrophy are all AR. In addition, the glycogen storage diseases for USMLE are all AR.

49. A 26-year-old man presents to the physician for a follow-up examination. He was diagnosed with a systolic murmur in middle school. Cardiac examination shows an S4 heart sound and paradoxical splitting of S2. There is no change in his murmur with Valsalva. Which of the following best describes this condition?
- A) Autosomal dominant; β -myosin heavy-chain gene
 - B) Autosomal dominant; bicuspid aortic valve
 - C) Autosomal recessive; β -myosin heavy-chain gene
 - D) X-linked dominant; β -myosin heavy-chain gene
 - E) X-linked recessive; β -myosin heavy-chain gene

The correct answer is B.

The diagnosis is bicuspid aortic valve. This is a familial condition inherited in an autosomal dominant pattern. The patient can develop aortic stenosis early in life (i.e., stenosis is not dependent on calcification later on).

Aortic stenosis demonstrates no change or improvement with less volume in the heart.

In contrast, hypertrophic obstructive cardiomyopathy (HOCM; mutations in β -myosin heavy-chain gene) gets worse with reduction in preload.

Valsalva is forced exhalation against a closed glottis. This results in increased intrathoracic pressure, decreased venous return, and decreased preload. Standing or sitting up from a supine position is another examination maneuver that will decrease preload.

Mid-systolic murmur that worsens with Valsalva / standing up $\rightarrow$ HOCM.

Mid-systolic murmur that improves or does not change with Valsalva / standing up $\rightarrow$ aortic stenosis.

An S4 heart sound signifies a stiffened left ventricle. This is due to increased afterload – i.e., from systemic hypertension or aortic valvular anomalies. On 2CK forms, an S4 can show up as right ventricular secondary to severe mitral stenosis (backing up through the lungs) or cor pulmonale.

Paradoxical splitting means A2 occurs after P2 (normally A2 occurs before P2). The later the valve closes, the greater the pressure in the respective ventricle. In the setting of severe left ventricular hypertrophy, A2 can occur so late that it actually closes on the opposite side of P2, hence paradoxical splitting. In right ventricular hypertrophy, P2 occurs later, so we get widened splitting. In atrial septal defects we get fixed (and wide) splitting.

Bottom line: Bicuspid aortic valve and HOCM are both autosomal dominant. HOCM gets worse with less preload; aortic stenosis demonstrates improvement or no change.

50. An African American couple comes to the physician for preconception counseling. On a previous visit, the man had a hemoglobin electrophoresis performed that showed the presence of HbA. The woman has not yet been tested. If 1 out of 13 African Americans have sickle cell trait, what is the probability this couple will have a son who has sickle cell trait?
- A) $1/13$
 - B) $1/26$
 - C) $1/34$
 - D) $1/48$
 - E) $1/52$
 - F) $2/13$

The correct answer is E.

Sickle cell anemia is an autosomal recessive disorder. The mutation is a glutamic acid → valine substitution on the β -globin gene. This results in more hydrophobic hemoglobin that is prone to sickling in states of dehydration and increased acidity.

Two mutations results in sickle cell disease, where HbS, rather than HbA (normal adult hemoglobin), is seen on hemoglobin electrophoresis.

One mutation results in sickle cell trait, where both HbA and HbS are seen on electrophoresis.

This man's hemoglobin electrophoresis only demonstrated HbA, indicating he neither has the disease nor is a carrier.

1/13 chance mom is a carrier

		A	a
Dad	A	AA	Aa
	A	AA	Aa

Since the carrier frequency in the African American population is 1 out of 13, the chance of this couple having a child with sickle cell trait is $1/13 \times 1/2 = 1/26$.

Therefore, the chance of having a son with sickle cell trait is $1/26 \times 1/2$ (chance of having son) = $1/52$.



YouTube
@mehlmanmedical



Instagram
@mehlman_medical



MEHLMANMEDICAL HY USMLE GENETICS

All material is copyrighted and the property of mehlmanmedical.

Copyright © [mehlmanmedical](#)